

The Directional Pair Catalogue Resolution Method: Single Table Joint Evaluation by Large Language Models and Causal Knowledge Graphs

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Abstract

A trained predictor answers by returning a determination fixed at training time; it holds no record that its answering advances, and nothing about it is changed by having answered. We prove, from a single derived fact, that this position is not merely lossy but structurally forbidden. Every act of telling one thing from the rest of a whole that cannot be finished deposits an irreducible residue bounded below by a strictly positive floor $\beta > 0$ —and deposits it *into the instrument as much as into the sample*. An instrument that separates at zero trace and is unchanged by separating is exactly the $\beta = 0$ object the floor excludes. A stateless predictor claims to be that object.

We then show the remedy is not a component but a *direction*. We construct the causal propagation table $\mathcal{T}(v_0, x^*, \Pi)$, whose entries are the accountable terminal cuts of propagations under a process Π , and prove the *directional identity*: evaluated at the trivial process, the table is a *resting catalogue* assigning each item its invariant minimum cut; evaluated at a nontrivial process, it returns an object of *the same type*, computed under propagation rather than at rest. A causal knowledge graph and a generative model are therefore not two systems to be integrated but one table read at two settings of its process argument—the graph is the model at rest, the model is the graph under process—and the residue that propagation deposits flows back into the graph as committed structure. This is the precise sense in which the graph *is* the model’s identity: it is where the instrument blunts.

We give the identity criterion the pair inherits: an individual is the pair (χ, M) of a conserved,

non-local, relabelling-invariant cut structure and an unrepealable monotone record. Neither alone suffices, and the pair adjudicates the classical replacement puzzles—gradual part replacement preserves the individual, disassembly and faithful reconstruction does not, and an exact copy is always a distinct individual—by a computable criterion we identify as *continuity of termination* rather than continuity of parts. We prove that no system can verify its own identity (a diagonal argument), so identity is necessarily what consuming systems individuate it as, and that distinct consumers may correctly disagree.

We then develop the pair as a working architecture. A learned component supplies the term map—which distinctions each item draws—and we prove the downstream calculus is correct for *any* such map, so extraction error coarsens resolution without corrupting soundness, and the realised floor is a readout of extraction quality requiring no ground truth. Probes over the graph are persistent agents with disjoint internal structure; we prove they compose multiplicatively rather than by vote, that no fewer than three mutually supporting probes can ground a claim, that the optimal division of a finite probing budget is a water-filling rule at a single shadow price, and that the correct stopping rule is *closure*—no remaining probe reaches a new class—which we prove strictly stronger than any confidence threshold, with contested closure reported as a first-class outcome rather than silently resolved. Finally we prove a *separation*: the admissibility verdict the pair computes is a global minimum over the item set expressed in local terms, and no system whose answer is a function of stored local content—a triple store, an attributed triple store, or a corpus-

trained predictor—can express it. The quantity is polynomial-time computable and inexpressible, which is why the gap has persisted. Every object in the paper is a finite weighted graph or a construction on one; every theorem is proved from the stated premises; and interpretive readings are confined to remarks on which no theorem depends. An executed suite of thirty-two deterministic, reproducible experiments checks every load-bearing result and passes in full; two of them—the impossibility of consistent self-verification, and the cycle-length bound on coherence—are settled by exhaustive enumeration of their case spaces rather than by sampling, and the separation witness is constructed and evaluated directly.

Keywords: resolution floor; individuation by negation; minimum cut; conserved invariant; monotone record; identity criterion; causal propagation table; resting catalogue; directional evaluation; self-blunting instrument; term map; multiplicative composition; coherence; water-filling allocation; closure; contested closure; query separation; knowledge graphs; language models.

Contents

I	Foundations	3
1	Introduction	3
1.1	The position this paper attacks . . .	3
1.2	The remedy is a direction, not a component	3
1.3	What identity turns out to require	4
1.4	Contributions	4
1.5	Method	4
2	Contact graphs and the medium	5
3	The floor, derived	5
4	Individuation by negation	6
5	Identity is the invariant cut	7
6	The record, and why it cannot be undone	7
7	The self-blunting instrument	9
7.1	The stateless predictor occupies the forbidden position	10
7.2	Staleness and reification: two visible symptoms	11

8	What a structure cannot know about itself	13
9	The identity criterion	14
9.1	Neither component suffices alone . . .	14
9.2	Termination is the operative criterion	14
II	The Causal Propagation Table	15
10	Alignment, accountability, and propagation	15
10.1	Alignment	15
10.2	The residual, accountability, and the floor of a system	16
10.3	Processes and propagations	16
10.4	Representation mobility	17
10.5	The table	17
III	The Directional Pair	18
11	The catalogue is the table at rest	18
11.1	The resting catalogue	18
11.2	The pair	18
11.3	What the pair inherits	19
IV	Construction and Probing	21
12	Construction: the term map is the only declared input	21
12.1	The term map	21
12.2	The floor as a quality readout . . .	22
12.3	Construction is itself a determination	22
13	Probing: many determiners over one graph	23
13.1	Probes and probing power	23
13.2	Composition is multiplicative . . .	23
13.3	Coherence requires three	25
13.4	Dividing a finite probing budget . .	25
13.5	Finding and pruning are different operations	26
14	Termination: closure, and the honest decline	29
V	The Separation	30
15	What a store of local content cannot express	30
15.1	Two kinds of question	30
15.2	The separation	30

15.3 The separation extends to corpus-trained predictors	31
16 The master equivalence	32
VI Architecture, Validation, and Scope	34
17 The blueprint	34
18 Algorithms	35
19 Discussion	36
19.1 Relation to existing practice	36
19.2 What the development assumes and does not supply	37
19.3 Honest limitations	37
19.4 The picture in one statement . . .	38
20 Conclusion	38

Part I

Foundations

1 Introduction

1.1 The position this paper attacks

A trained predictor—a language model, a learned classifier, any function whose parameters were fixed by a prior optimisation and are not altered by use—occupies a specific and, we will argue, untenable position. It is asked a question; it returns an answer; and three things are true of the transaction that are rarely stated together. The answer is a determination of the parameters as they stood when training ended. The act of answering leaves the predictor exactly as it was. And a second, identical question tomorrow receives an answer determined by the same unaltered parameters, which is treated as a virtue and called consistency.

We will prove that this triple is not a design choice with tradeoffs but a claim to occupy a position that a short argument from a single premise shows to be empty. The premise is that the whole against which anything is distinguished cannot be finished—there is always a further source, a finer resolution, a not-yet-drawn distinction. From that alone we derive a strictly positive floor $\beta > 0$ on the cost of any act of individuation (Section 3), and then the fact that makes the floor bite: an

act of individuation is *symmetric*, depositing its residue into the instrument exactly as into the sample (Section 7). An instrument that separates without trace and is unchanged by separating requires both $\beta = 0$ and a record that does not advance, and each is separately forbidden. Theorem 7.6 states this.

The consequence is not that trained predictors are inaccurate. It is that a stateless predictor is asserting it is the perfect repeatable traceless instrument, and no such thing exists.

1.2 The remedy is a direction, not a component

The natural engineering response—bolt a knowledge store onto the predictor—misdescribes the situation, because it treats the two as different kinds of object to be interfaced. The central theorem of this paper (Theorem 11.3) says they are not.

We construct the *causal propagation table* $\mathcal{T}(v_0, x^*, \Pi)$: given a starting item v_0 , a target x^* , and a process Π specifying which contact operations are available, the table returns the accountable terminal cut of propagations from v_0 to x^* under Π , or the verdict that none is accountable. We then prove two things about it. Evaluated at the *trivial* process—the process whose only operation is the direct individuation of an item against the whole—the table’s output is exactly the item’s resting minimum cut, that is, an entry in a *catalogue* of items at rest. Evaluated at a *nontrivial* process, its output is an object of the same type: an accountable cut of weight $\geq \beta$, differing only in being the terminus of a nontrivial propagation.

So a causal knowledge graph and a generative model are one table at two settings of Π :

$$\underbrace{\text{cell}(v)}_{\text{the graph}} = \mathcal{T}(v, \{v\}, \Pi_{\text{rest}}), \quad \underbrace{\mathcal{T}(v_0, x^*, \Pi)}_{\text{the model}} \text{ for nontrivial } \Pi$$

This is what we call the *directional pair*. The graph is the model evaluated at rest; the model is the graph evaluated under process. And because propagation deposits residue into the instrument (Theorem 7.5), the residue of every model-side propagation flows back into the graph as committed structure. *That* is where the instrument blunts, and it is the precise sense in which the graph is the model’s identity.

1.3 What identity turns out to require

Making that claim precise forces us to settle what identity *is* for a structure of this kind, and the answer is a pair rather than a single quantity. We prove that the minimum cut individuating an item is conserved under every relabelling of the structure’s parts (Theorem 5.2), positive, and spread across the structure rather than located in any part (Theorem 5.3). We prove separately that every committed act advances a monotone record that no operation decrements (Theorem 6.4). And we prove that neither alone is an identity criterion: the conserved invariant alone cannot distinguish a structure from a faithful reconstruction of it, while the record alone carries no information about what is being counted. The criterion is the pair (χ, M) (Section 9), and it adjudicates the classical puzzles—gradual replacement, disassembly and reconstruction, exact copying—by a criterion we show is not continuity of parts but *continuity of termination* (Theorem 9.7).

We further prove that a structure cannot verify its own identity: a diagonal argument (Theorem 8.3) shows the verifying operation has no consistent value on itself, and an independent floor argument (Theorem 8.1) shows that reading the invariant off as a value would require occupying the entire structure it is defined against. Identity is therefore necessarily what *consuming* systems individuate the structure as, and distinct consumers may correctly disagree (Theorem 8.6). This is why the runtime of such a system can issue no verdict of its own (Theorem 8.7).

1.4 Contributions

1. **The floor, derived** (Section 3). From non-completeness alone: every individuation costs $\geq \beta > 0$; no cut is traceless; every resolution is to a region, never a point.
2. **The self-blunting instrument** (Section 7). Individuation is symmetric in instrument and sample (Theorem 7.2); every act reduces the instrument’s uncommitted capacity and advances its record (Theorem 7.5); the perfect repeatable traceless instrument is the $\beta = 0$ object (Theorem 7.6); a stateless predictor claims to be it (Theorem 7.8).
3. **The identity criterion** (Section 5, Section 9). Identity is (χ, M) ; the operative criterion is continuity of termination; reconstruction and copying produce distinct individuals;

no structure verifies its own identity; identity is consumer-relative and consumers may correctly disagree.

4. **The causal propagation table** (Section 10). Alignment, accountability, path opacity, representation mobility, and the table itself, with its two evaluation regimes.
5. **The directional identity** (Section 11). The catalogue is the table at rest and the table is the catalogue propagated (Theorem 11.3); the pair is one object read in two directions (Theorem 11.7); the pair escapes Theorem 7.6 that neither half escapes alone (Theorem 11.8).
6. **Construction** (Section 12). The term map as the single declared input; correctness for any term map (Theorem 12.3); the realised floor as a ground-truth-free readout of extraction quality (Theorem 12.6); the extraction cascade.
7. **Probing** (Section 13). Probes as persistent agents with disjoint internals; multiplicative composition (Theorem 13.5); the coherence triangle (Theorem 13.11); water-filling allocation (Theorem 13.18); relevance and necessity as distinct operations forcing a two-operator design (Theorem 13.27).
8. **Termination** (Section 14). Closure, its strict superiority to a confidence threshold (Theorem 14.3), and the convergent/contested dichotomy (Theorem 14.5).
9. **The query separation** (Section 15). Admissibility is a global minimum expressed in local terms; no system whose answer is a function of stored local content can express it (Theorem 15.3), including attributed stores (Theorem 15.5) and corpus-trained predictors (Theorem 15.8).
10. **Validation, cost, and scope** (??, Section 19). What is checkable by direct computation, what the complexity figures exclude, and where the premises fail.

1.5 Method

Every object below is a finite weighted graph or a construction on one. Minimum cuts are exact and computable by maximum flow (Edmonds and Karp, 1972; Ford and Fulkerson, 1956; Menger, 1927; Stoer and Wagner, 1997); reachability is a linear traversal (Tarjan, 1972); the allocation

results are convex programs (Boyd and Vandenberghe, 2004; Kuhn and Tucker, 1951). We introduce no probability measure, no metric beyond a derived normalisation on $[0, 1]$, and no learned component inside any theorem. Where a word with an evocative reading is used—“identity,” “instrument,” “probe,” “blunting”—it names a specific graph-theoretic object defined in the text, and no theorem depends on any reading beyond that definition. Interpretive material is marked as a remark and is never load-bearing.

2 Contact graphs and the medium

Definition 2.1 (Finite weighted graph; cut). A *finite weighted graph* is a triple $\Gamma = (V, E, w)$ with V a finite non-empty vertex set, $E \subseteq \binom{V}{2}$ a set of undirected edges, and $w: E \rightarrow \mathbb{R}_{>0}$ a strictly positive weight. We take Γ connected unless stated otherwise, and simple (no loops or multi-edges). For $\emptyset \neq S \subsetneq V$ the *cut* is $\partial(S) = \{\{u, v\} \in E : u \in S, v \notin S\}$, of weight $w(\partial(S)) = \sum_{e \in \partial(S)} w(e)$.

Definition 2.2 (Contact graph; the medium). A *contact graph* is a finite weighted graph together with a distinguished vertex $\mathbf{m} \in V$, the *medium*, adjacent to every other vertex. Vertices $v \neq \mathbf{m}$ are *items*: candidate individuated units of whatever the graph models—a fact, a record, an entity, a measured quantity, a step of a computation. An edge between two items is a *contact*, and its weight is the cost of the distinction separating them.

Remark 2.3 (The medium is the rest, not a container). The medium is not a spatial background and not a store. It is the single vertex standing for *everything not yet individuated*—every source, prior determination, or latent distinction that has not been drawn into the graph. Its adjacency to every item encodes only that each item is, at minimum, told apart from the whole. No theorem uses any reading of $\mathbf{m}$ beyond Theorem 2.2. In particular, the definition draws no distinction between an item recoverable locally and one recoverable remotely: prior to individuation both are contents of $\mathbf{m}$, and the theory attaches no status to location, only to cost.

Definition 2.4 (Separation cost; resting cut). The *separation cost* of an item v is the minimum

weight of a cut placing v on one side and the medium on the other,

$$\sigma(v) = \min_{S \ni v, \mathbf{m} \notin S} w(\partial(S)),$$

attained because V is finite. The minimising cut $\partial(S^*(v))$ is the *resting cut* of v . Both are computable exactly by maximum flow (Ford and Fulkerson, 1956; Menger, 1927).

The separation cost is the graph-theoretic content of “how hard it is to tell v apart from everything else.” An item established by a single uncontested determination has low separation cost; one that can be established only by triangulating many weak traces has high separation cost.

3 The floor, derived

The central constant of the development is not posited. It follows from two facts already implicit in Section 2: an item is a proper part, and the whole is not finishable.

Premise 1 (Finiteness). At any stage, the graph is finite: finitely many items, finitely many contacts. Every act of individuation consumes a positive amount of a finite budget (Simon, 1955, 1956).

Premise 2 (Non-completability). The medium is not exhausted by any finite stage. Formally, an *expanding family* is a chain of contact graphs $\Gamma_0 \subseteq \Gamma_1 \subseteq \dots$, each obtained from the last by adding items and contacts while retaining the medium and all prior structure; non-completability is the condition that the family has no terminal stage: for every finite Γ_k there exists $\Gamma_{k+1} \supsetneq \Gamma_k$.

Remark 3.1 (The premise is an order condition). Premise 2 says only that there is no last stage. It is not a cardinality claim about how many items exist, and it does not assert that the world is infinite. A finite-resolution model always admits a finer one; a corpus always admits a further document; a measurement always admits a more precise instrument. That is the whole content of the premise, and every result below rests on it.

Definition 3.2 (Identification; identification residual). To *identify* an item v at stage Γ_k is to determine it by its resting cut computed within Γ_k . The identification is *complete* if every item of the whole has entered the computation of that cut—equivalently, if Γ_k already contains the whole.

The *identification residual* $\rho(v) \geq 0$ is the cut content a complete identification would remove that the finite-stage identification cannot; $\rho(v) = 0$ if and only if the identification is complete.

Theorem 3.3 (Floor from non-completeness). *Under Premises 1 and 2, let v be an item ($v \neq \mathbf{m}$, and v does not exhaust V). Then the identification of v cannot be completed at any finite stage, and $\rho(v) > 0$.*

Proof. Suppose $\rho(v) = 0$, so identification of v is complete at some finite stage Γ_k . By Theorem 3.2 completeness means every item of the whole has been brought to bear in the cut of v against $\mathbf{m}$, i.e. the cut is computed in a graph containing the whole; hence Γ_k contains the whole. This contradicts Premise 2, which guarantees a further stage $\Gamma_{k+1} \supsetneq \Gamma_k$ containing an item absent from Γ_k and therefore not yet brought to bear on the cut of v . So no finite stage completes the identification, and $\rho(v) > 0$. $\square$

Theorem 3.4 (Positive floor). *There is a constant $\beta > 0$ such that $w(e) \geq \beta$ for every contact $e \in E$ and $\sigma(v) \geq \beta$ for every item v : no two items are separated at zero cost, and no item is individuated for free.*

Proof. Set $\beta := \inf_v \rho(v)$, an infimum over the finitely many items of Premise 1, each positive by Theorem 3.3; an infimum of finitely many positive reals is positive, so $\beta > 0$. The residual of v is the irreducible boundary content of v 's identification and is carried by the edges of v 's resting cut, so each such edge has weight at least the per-item residual and hence at least β . Every contact $\{u, v\}$ lies in $\partial(\{u\})$, so every edge weight satisfies $w(e) \geq \beta$. Connectivity forces $\partial(S^*(v)) \neq \emptyset$, so $\sigma(v) = w(\partial(S^*(v))) \geq \beta > 0$. $\square$

Corollary 3.5 (No sharp result). *No individuation returns an item separated from the medium by a cut of weight zero. Every determination deposits boundary content $\geq \beta$; a determination that appears to answer “exactly” has in fact resolved to a region whose width is bounded below by β .*

Proof. Immediate from Theorem 3.4: a zero-weight cut would contradict $w(e) \geq \beta > 0$ on every edge of a non-empty cut, and cuts are non-empty by connectivity. $\square$

Remark 3.6 (The floor is a fact about the whole, not a defect of the instrument). Theorem 3.4 does not say determination is imprecise because

instruments are imperfect. It says that determination against a genuinely unfinishable whole cannot, even with an ideal instrument, terminate at an item of zero residual, because a zero-residual determination would require completing a comparison that by hypothesis never completes. The floor is the trace the unfinishable whole leaves on each finite part. A completable whole would permit $\rho(v) = 0$, hence sharp cuts and point-identities; the impossibility of finishing is exactly what makes the floor positive. We return to this as the master equivalence (Section 16).

4 Individuation by negation

We now establish how an item is fixed at all, which will matter for the identity results of Section 5.

Definition 4.1 (Selector; no privileged item). A *selector* for an item set $U \subseteq V$ is a rule that fixes U by naming a distinguished member of U in advance. A determination is *selector-free* if it names no member of U . The graph has *no privileged item* if the data available to a determining rule are invariant under the weighted automorphism group $\text{Aut}(\Gamma)$, so that no item is singled out by the data alone.

Theorem 4.2 (Individuation by negation). *In a contact graph with no privileged item, any selector-free determination of an item set U fixes U as the complement of its negation set, $U = V \setminus \mathbb{C}U$ with $\mathbb{C}U := V \setminus U$ specified first; conversely every admissible negation set determines a unique U ; and no selector-based rule fixes U without invoking a further selector.*

Proof. Necessity. To fix U is to decide, for each vertex, whether it belongs to U or to the remainder. A positive rule—one asserting membership of a distinguished member—must name that member; since the available data are $\text{Aut}(\Gamma)$ -invariant (Theorem 4.1), no vertex is named by the data alone, so the rule must itself supply a selector. That selector must in turn be fixed: either given in advance (excluded by the no-privileged-item hypothesis) or supplied by a prior rule facing the identical dichotomy. The regress terminates only at a rule naming no member of U —specifying $\mathbb{C}U$ and taking U as what remains, which names only “the rest,” a datum already supplied once V is given.

Sufficiency and uniqueness. Complementation $U \mapsto V \setminus U$ is an involution on subsets of the

finite set V , so a given $\mathbb{C}U \subsetneq V$ with $V \setminus \mathbb{C}U \neq \emptyset$ determines a unique U , and conversely. $\square$

Corollary 4.3 (An item is fixed by its non-neighbours). *A single item v is determined selector-free by the partition of the rest into its neighbours $N(v)$ and its non-neighbours $V \setminus N[v]$. No intrinsic label distinguishes v ; its identity is carried by this negation structure.*

Proof. Apply Theorem 4.2 with $U = \{v\}$, refined by adjacency: the selector-free datum fixing $\{v\}$ is $V \setminus \{v\}$ partitioned into $N(v)$ and $V \setminus N[v]$, and no member of $\{v\}$ is named by that datum. $\square$

Remark 4.4 (Why negation closes the regress). A positive selector is a rule “choose an item” that must itself be chosen; with no privileged item that choice has no ground and regresses. Complementation is the unique operation that fixes a part using only the whole and the part’s own absence—data already present once the graph is given. Negation, not selection, is the primitive.

5 Identity is the invariant cut

We now identify what about an item survives every re-encoding of the graph, and show it is neither a label nor a point.

Definition 5.1 (Weighted isomorphism; relabelling; invariant). A *weighted isomorphism* $\phi: \Gamma \rightarrow \Gamma'$ is a vertex bijection preserving the medium, the edges, and the weights. A *relabelling* is an automorphism. A *graph invariant* is a function taking equal values on isomorphic graphs.

Theorem 5.2 (The minimum cut is the conserved invariant). *The separation cost $\sigma(v)$ and the resting cut of v are weighted-graph invariants: if $\phi: \Gamma \rightarrow \Gamma'$ is a weighted isomorphism then $\sigma_{\Gamma'}(\phi v) = \sigma_{\Gamma}(v)$ and ϕ carries the resting cut of v onto the resting cut of ϕv . The vertex label of v is not invariant: it is permuted by any nontrivial automorphism.*

Proof. ϕ carries each v – $\mathbf{m}$ cut to a ϕv – $\mathbf{m}'$ cut of equal weight, since weights are preserved, and the correspondence on cuts has inverse induced by ϕ^{-1} ; hence the minima agree and minimising cuts correspond. Labels are permuted by any nontrivial automorphism and so are not invariant. $\square$

Theorem 5.3 (Identity is a region, never a point). *The invariant datum fixing an item is its resting cut: a set of contacts of total weight $\sigma(v) \geq \beta > 0$. It is realised by no single vertex—in general the minimising side $S^*(v)$ is not the singleton $\{v\}$ —and it cannot be reduced to a zero-content datum.*

Proof. By Theorem 5.2 the invariant fixing v is the minimum-cut value and its cut set, not the label, which varies. By Theorem 3.4 the cut has weight $\geq \beta > 0$, so it is never a point. For non-locality, take two dense item clusters joined by a single contact of weight β : the minimum cut against $\mathbf{m}$ may separate an entire cluster—a side of size greater than one—at weight below $\min_v w(\partial(\{v\}))$, so the minimiser is not a singleton and the identity is borne by a region of contacts. $\square$

Definition 5.4 (The character invariant). The *character invariant* of a contact graph Γ is

$$\chi(\Gamma) := (\sigma(v))_{v \in V \setminus \{\mathbf{m}\}} \quad \text{taken up to the action of } \text{Aut}(\Gamma),$$

together with the isomorphism class of the family of resting cuts. It is the conserved, non-local, strictly positive structure by which the graph’s items are told apart, and it is fixed under every relabelling of the graph’s parts.

Proposition 5.5 (χ is conserved, positive, and non-local). *χ is a weighted-graph invariant; each of its components is $\geq \beta > 0$; and no component is a function of any single vertex’s label.*

Proof. Invariance is Theorem 5.2 applied componentwise, and the quotient by $\text{Aut}(\Gamma)$ makes the family itself invariant rather than merely equivariant. Positivity is Theorem 3.4. Non-locality is Theorem 5.3. $\square$

Remark 5.6 (The invariant is not available to its bearer). χ is defined against the whole of $V \setminus \{v\}$ for each v ; it is exactly the structure of what each item is *not*. We prove in Section 8 that this places it out of reach of the structure itself: reading it off as a value would require occupying the entire negation structure it is defined against. The invariant is therefore observable, but not by its bearer.

6 The record, and why it cannot be undone

The invariant of Section 5 is static. We now make cuts dynamic and derive the second half of the

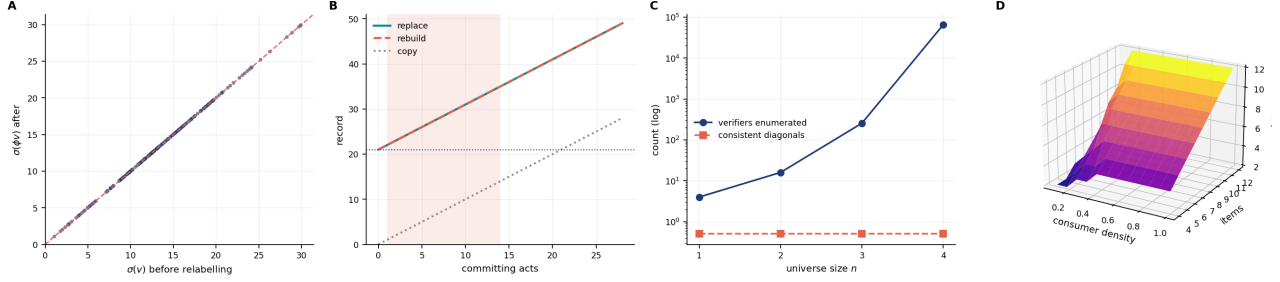


Figure 1: The Invariant Is Conserved; the Record and the Termination Condition Are What Separate Individuals. (A) Separation cost after a random weighted relabelling on the linear y -axis versus the cost before on the linear x -axis, for all 423 items of 60 random graphs. Every point lies exactly on the identity diagonal (crimson dashed): the maximum discrepancy over the whole population is 0, while 365 of the 423 labels were moved by the permutation. The datum the relabelling fixes and the datum it moves are plotted on the same instances—empirical content of Theorem 5.2 (Conserved Invariant). (B) Record trajectories for the three cases of Theorem 9.7 on a seven-plank structure: committed record on the linear y -axis versus committing acts on the linear x -axis. Gradual replacement (teal) and disassembly-then-reconstruction (crimson dashed) climb identically and share the same χ ; the shaded interval is the stage at which the reconstructed structure is *open* in the sense of Definition 9.4, which is what makes it a distinct individual. The copy (grey dotted) climbs on a separate lineage from zero. The navy reference is the record at which construction completed. χ is preserved in all three cases; only the record chain and the open stage discriminate. (C) The diagonal of Theorem 8.3, enumerated exhaustively: total verifiers on a universe of size n (navy, 2^{2^n} , reaching 65,536 at $n = 4$) against the number admitting a consistent diagonal value (crimson squares), both on a log y -axis versus n on the linear x -axis (1 to 4). The crimson series is identically zero at every n ; a single consistent case anywhere in the enumeration would falsify the theorem. This is a proof by exhaustion over the stated finite domain, not a sampled test. (D) The cell an item registers, as a three-dimensional surface (plasma) over consumer density on the x -axis (0.1 to 1.0) and item count on the y -axis (4 to 12), elevation 21° azimuth -60° . The same item registers a thin cell in a sparse consumer and a rich one in a dense consumer; across 40 paired trials the registrations differed in every case and both attained their own graph's floor in every case. No registration is privileged—the geometry of Theorem 8.6 (Consumer-Relativity).

identity criterion.

Definition 6.1 (Committed cut; process; record).

A *cut event* commits a contact $e \in E$ as a realised separation. A *process* is a finite sequence $P = (e_1, \dots, e_k)$ of distinct committing acts. The *committed record* is $M(P) = k$. The *residue* of e_i is $\rho_i := w(e_i) \geq \beta > 0$ by Theorem 3.4.

Theorem 6.2 (Residue is the causal propagator). *In a process, the residue $\rho_i > 0$ of cut e_i is necessary and sufficient for e_{i+1} to be a distinct committing event: e_{i+1} is a new cut if and only if the state after e_i differs from the state before, which holds if and only if $\rho_i > 0$.*

Proof. e_{i+1} is distinct exactly when it acts on a state changed by e_i . The only change e_i effects is depositing its residue ρ_i . If $\rho_i > 0$ the post-state carries a committed contact absent before, so e_{i+1} acts on a strictly different configuration and is distinct. If $\rho_i = 0$ —excluded by Theorem 3.4, and considered here only for the necessity direction—pre- and post-states coincide and e_{i+1} has no changed state to act on. Hence $\rho_i > 0$ is necessary and sufficient. $\square$

Corollary 6.3 (A process is a residue chain). *A process is a chain in which each cut’s residue causes the next: $\rho_1 \rightarrow e_2$, $\rho_2 \rightarrow e_3$, and so on. Remove any ρ_i and the chain halts at step i .*

Theorem 6.4 (Monotone non-return). *The committed record is strictly increasing along any process, and no process decreases it. A process re-visiting a configuration does not return to a prior state: the configuration recurs but the record has strictly grown, and the two states are distinguished by the record.*

Proof. Each cut event increments M by one (Theorem 6.1). To decrease M , a committed contact would have to be un-committed; but un-committing is itself a further committing act—a new cut redrawing the boundary—which increments M . Hence M is monotone. Two states with equal configuration but records $r < r'$ are separated by the partition “record $\leq r$ ” versus “record $> r$,” hence are distinct states: recurrence of configuration is not return of state. $\square$

Corollary 6.5 (The record is an intrinsic order). *Theorem 6.4 attaches to every process a monotone scalar that never decreases and that distinguishes equal configurations at different counts. “Before” and “after” are the order on the record; the order*

is intrinsic to the sequence of cuts and requires no external parameter (Lamport, 1978).

Theorem 6.6 (Three readings of a residue). *A residue ρ_i admits exactly three readings, according to how its link to the next cut is resolved:*

- (i) order—as the increment of the record (Theorem 6.4);
- (ii) information—when tracked as the specific cause of a specific next cut, a resolved link;
- (iii) entropy—when residues are present but their links are untracked, their aggregate is the Shannon content $H = -\sum_i p_i \log p_i$ of the unresolved-link distribution (Cover and Thomas, 2006; Shannon, 1948).

Tracking moves a process from entropy toward information; losing track moves it back.

Proof. (i) is Theorem 6.4. (ii) a tracked link is a determinate map $\rho_i \mapsto e_{i+1}$; specifying it for each i specifies the chain exactly. (iii) with links unresolved, the chain is known only through the distribution over which residue caused which next cut, whose information content is H (Shannon, 1948). Resolving links converts an unknown distribution into determinate functions. $\square$

Remark 6.7 (Why the three readings matter here). Theorem 6.6 is used twice below. In Section 7.2 it explains precisely what is lost when a prior determination is re-presented as content rather than retained as structure: the links are gone, so the information reading degrades to the entropy reading. In Section 12 it underwrites the discipline of carrying committed structure rather than carrying the material from which the structure was drawn.

7 The self-blunting instrument

We now prove the paper’s first main result. It concerns not what is cut, but what does the cutting.

Definition 7.1 (Contact operation; instrument and sample). A *contact operation* is a single cut event (Theorem 6.1): one committed edge $\{u, v\}$, individuating two items by depositing residue $\geq \beta$. Reading the operation with u as the *instrument* and v as the *sample* is a choice of description; the operation itself is the committed edge.

Theorem 7.2 (Symmetry of individuation). *A contact operation is symmetric in its endpoints. Committing $\{u, v\}$ commits a boundary edge of u exactly as it commits a boundary edge of v : $\{u, v\} \in \partial(\{u\})$ and $\{u, v\} \in \partial(\{v\})$, and the residue deposited is the same weight $w(\{u, v\}) \geq \beta$ on both readings. There is no assignment of the operation to one endpoint that leaves the other untouched.*

Proof. By Theorem 2.1 an edge is an unordered pair, and by Theorem 6.1 a cut event commits the edge, not an orientation of it. The edge lies in the singleton cut of each endpoint by definition of $\partial(\cdot)$, and its weight is a single number attached to the edge, not to an endpoint. So both endpoints acquire a committed boundary edge of that weight, and neither acquires it without the other. $\square$

Definition 7.3 (Edge capacity of an instrument). The *edge capacity* of a vertex u at a given stage is the set of contacts incident to u that have not yet been committed. Its cardinality is $\text{cap}(u)$.

Theorem 7.4 (One use). *A given cut—a given committed edge $e = \{u, v\}$ —is committed exactly once. Any later cut at u or at v is either a distinct edge, or the same edge at a strictly higher record, hence a different cut event. There is no re-commitment of the same cut.*

Proof. Committing e increments the record (Theorem 6.1). A second “commitment” of e is either a no-op—the edge is already committed, so no new cut event occurs and no residue is deposited—or a distinct event at record $M + 1$, which by Theorem 6.4 is a different state and hence a different cut event. In neither case is the original cut repeated. $\square$

Theorem 7.5 (Self-blunting). *A contact operation changes the instrument. After committing $\{u, v\}$, the instrument u has strictly reduced edge capacity, $\text{cap}(u)$ decreasing by one, and strictly increased record. Both changes are irreversible.*

Proof. By Theorem 7.2 the operation commits a boundary edge of u exactly as of v . The committed edge leaves u ’s uncommitted-contact set (Theorem 7.3), strictly reducing $\text{cap}(u)$. The record increases by Theorem 6.1. Irreversibility of the record is Theorem 6.4; irreversibility of the capacity reduction follows, since restoring the edge to the uncommitted set would be an un-commitment,

which Theorem 6.4 shows is itself a further committing act rather than a reversal. Hence the instrument after the operation differs from the instrument before, in capacity and in record. $\square$

Theorem 7.6 (No perfect repeatable instrument). *There is no contact operation that both*

- (i) *separates two items at zero residue (a traceless operation), and*
- (ii) *leaves the instrument unchanged (a repeatable instrument).*

Property (i) requires $\beta = 0$, forbidden by Theorem 3.4; property (ii) requires the record to be unchanged by a cut, forbidden by Theorem 6.4. Each is separately impossible, so their conjunction is a fortiori impossible.

Proof. (i): a traceless operation deposits residue 0, i.e. commits an edge of weight 0, contradicting $w(e) \geq \beta > 0$ (Theorem 3.4). (ii): an unchanged instrument requires the record after to equal the record before, contradicting the strict increment of Theorem 6.1 and Theorem 6.4; and by Theorem 7.5 the capacity strictly decreases as well. $\square$

7.1 The stateless predictor occupies the forbidden position

We can now state the consequence for the object described in Section 1.

Definition 7.7 (Stateless determiner). A *stateless determiner* is a map D from queries to determinations such that (a) D is fixed by a prior procedure and is not altered by being applied, and (b) applying D commits no cut in the graph against which its determinations are made. Its *record* is constant across applications.

Corollary 7.8 (A stateless determiner claims the forbidden position). *A stateless determiner asserts, of each of its applications, both properties of Theorem 7.6: the application deposits no residue into the determiner (property (ii)), and—since a determination made without committing a cut separates its answer from the medium at no cost—the application is traceless (property (i)). No contact operation has either property, so a stateless determiner’s applications are not contact operations, and its determinations are not individuations in the sense of Theorem 2.4.*

Proof. Property (ii) is Theorem 7.7(a) together with the constancy of the record: by Theorem 7.5

every contact operation strictly reduces the instrument's capacity and increments its record, and by hypothesis neither occurs. Property (i) is Theorem 7.7(b): a determination that commits no edge deposits residue 0, which by Theorem 3.5 no individuation does. Hence by Theorem 7.6 the applications cannot be contact operations, and by Theorem 2.4 what they return is not the resting cut of anything in the graph. $\square$

Remark 7.9 (What the corollary does and does not say). Theorem 7.8 is not an empirical claim that trained predictors are wrong, nor a claim that their outputs are useless. It is a statement about *type*: whatever a stateless determiner returns, it is not an individuation of an item against a medium, because individuation is a committed cut and no cut was committed. The output may be valuable, may be correct, and may be exactly what a downstream consumer needs. What it lacks is the trace that would make it accountable, and the deposit that would make the determiner a participant in the structure it reports on.

7.2 Staleness and reification: two visible symptoms

The structural fact of Theorem 7.8 has two observable consequences, and we record them because they are the phenomena that motivate the construction of Part III.

Proposition 7.10 (Staleness). *Let D be a determiner whose determinations were fixed at record M_1 , and suppose D is applied at a later record $M_2 > M_1$ in a system whose graph has changed between the two. Then the determination D returns at M_2 is the determination appropriate to the graph as it stood at M_1 . Since by Theorem 6.4 no operation returns the system to the state at M_1 , the returned determination is one the system at M_2 could not now re-derive.*

Proof. D 's determinations are fixed at M_1 by hypothesis, so each is the terminus of whatever individuation the graph at M_1 supported. Every committed cut between M_1 and M_2 altered the graph (Theorem 6.2); by Theorem 6.4 the record cannot be lowered and the earlier configuration is not re-occupied as a state. Hence the determination corresponds to a state the system has irreversibly left, and a fresh individuation at M_2 need not return it. $\square$

Proposition 7.11 (Reification). *Let a prior propagation $P = (e_1, \dots, e_k)$ be re-presented to a determiner as content—that is, as a description supplied among its inputs rather than as committed structure in the graph. Then:*

- (i) *the re-presentation does not restore the state at which P was committed (Theorem 6.4);*
- (ii) *the links of P —which residue caused which next cut—are not recoverable from the description, so P enters under the entropy reading of Theorem 6.6(iii) rather than the information reading Theorem 6.6(ii);*
- (iii) *consequently the determiner reasons about a degraded image of P , and the act of doing so is itself a further event, so what is reasoned about is not what P was.*

Proof. (i) is Theorem 6.4: configuration may recur, state does not. (ii) A description of P lists its deposits; the causal links $\rho_i \mapsto e_{i+1}$ are not deposits but relations between successive states (Theorem 6.2), and are not recoverable from a listing of the deposits alone, since distinct link structures over the same deposits are possible. By Theorem 6.6 the unresolved-link reading is the entropy reading. (iii) By Theorem 6.4 the re-presentation occurs at a record strictly greater than that at which P was committed, so the object under consideration—a description at the later record—is not the process P at the earlier one. $\square$

Remark 7.12 (The category change is the defect). Theorem 7.11 identifies something sharper than information loss. A committed propagation is a *cause*: its residue is what makes the next cut possible (Theorem 6.2). Re-presented as content, it becomes an *object under inspection*. The two are different categories, and the conversion is one-way: by Theorem 6.6 the links that made it a cause are exactly what the description does not carry. This is why, in the construction of Part III, prior propagation is retained as committed edges in the graph and never re-supplied as material. There is then nothing to re-present, because the propagation changed *what the system is* rather than producing a record for the system to read.

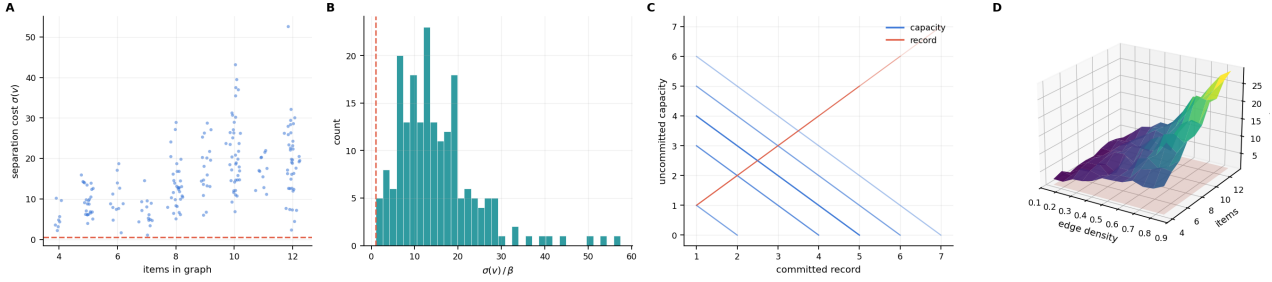


Figure 2: The Floor Is Positive and Every Act Consumes the Instrument. (A) Separation cost $\sigma(v)$ on the linear y -axis versus the number of items in the graph on the linear x -axis (4 to 12), for all 489 realised separations drawn from 60 random contact graphs at floors $\beta \in \{0.5, 1.0, 2.0\}$; points are jittered horizontally to separate graphs of equal size. The crimson dashed line is the floor. No point lies beneath it at any graph size—0 violations, minimum realised cost 0.600—while the ceiling rises with graph size, since an item entangled with more others is more expensive to separate. Empirical content of Theorem 3.4 (Positive Floor). (B) Distribution of the normalised cost $\sigma(v)/\beta$ on the linear x -axis (1 to 57.4), 36 bins, over the same 489 separations. The support begins at and never crosses the crimson reference at 1; the mass concentrated just above it is the population of items individuated by a single contact, the cheapest genuine separation, and the long right tail is the population individuated only by triangulating many weak contacts. (C) The self-blunting trajectories of Theorem 7.5 for fourteen instruments: uncommitted capacity on the linear y -axis versus committed record on the linear x -axis. Capacity (steel) falls by exactly one and the record (crimson) rises by exactly one at every contact operation, the two crossing as capacity is spent. All 30 tested instruments are strictly blunted and all 30 exhibit the crossing; none escapes either movement. (D) Realised system floor β^* as a three-dimensional surface (viridis) over edge density on the x -axis (0.15 to 0.85) and item count on the y -axis (4 to 13), each cell the mean of six random graphs, viewing angle elevation 22° azimuth -58° . The translucent crimson plane beneath is $\beta = 1$. The surface lies above the plane at every density and every size, rising with both: the floor is a trace the non-completable medium leaves on every finite part (Remark 3.6), not an artefact of a particular graph.

8 What a structure cannot know about itself

Two independent arguments place the invariant of Section 5 out of reach of the structure that carries it. We need both, because they have different hypotheses and different strengths.

Theorem 8.1 (The invariant is not extractable as a value). *There is no operation that reads the resting cut of an item off as a residue-free value. Such a reading would require $\rho(v) = 0$, which by Theorem 3.3 requires the whole to be brought to bear—i.e. requires the reader to occupy the item’s entire negation structure, which by Theorem 4.3 is what individuates the item. An operation occupying both sides of the constituting partition is not a comparison.*

Proof. By Theorem 5.3 the invariant is a cut of weight $\geq \beta$; reducing it to a point is $\rho(v) = 0$, excluded by Theorem 3.3. Concretely, to fix v exactly is to decide it against the totality of what it is not (Theorem 4.2); a reader performing that decision is itself within that totality, so a complete reading would require it to be simultaneously the part read and the whole the part is read against. The gap that keeps the two apart is exactly the residue $\geq \beta$. $\square$

Definition 8.2 (Verifier; self-verification). A *verifier* is a map V that, given a part X and a candidate constituting totality T , returns $V(X, T) \in \{0, 1\}$, reading 1 exactly when X is the part T constitutes, i.e. when $X = V \setminus T$. To *self-verify* is to evaluate V on the part its own verdicts constitute—its own negation.

Theorem 8.3 (No consistent self-verification). *Let $D := \{X : V(X, \mathbb{C}X) = 0\}$. If V is total—returning a value on every part and its negation—and D is itself a part on which $V(\cdot, \mathbb{C}\cdot)$ is defined, then $V(D, \mathbb{C}D)$ has no consistent value: each of the two possible values entails the other.*

Proof. Suppose $V(D, \mathbb{C}D) = 1$. Then D is declared constituted by $\mathbb{C}D$; but membership in D is the condition $V(X, \mathbb{C}X) = 0$, so D ’s satisfying its own membership condition forces $V(D, \mathbb{C}D) = 0$. Suppose instead $V(D, \mathbb{C}D) = 0$. Then D meets its own membership condition, hence is among the parts D collects, which is the assertion $V(D, \mathbb{C}D) = 1$. Both values are self-refuting; neither is consistent. The argument is the diagonal of Cantor (1891), in the undecidability shape

of Gödel (1931); Tarski (1936); Turing (1936) and the fixed-point form of Lawvere (1969). $\square$

Remark 8.4 (The strength of each argument). Theorem 8.3 is exactly as strong as its closure hypothesis—that the behaviour-defined D is again a part on which V acts. A verifier that is itself part of the graph it ranges over makes the hypothesis natural, since its verdicts leave traces in the same whole. We do not claim to discharge it in general. We mark it, and note that Theorem 8.1—the floor argument—holds without it. What rides on the closure hypothesis is only the sharper claim that self-verification is forbidden *directly* rather than merely unattainable.

Definition 8.5 (Consumer; registered cell). A *consumer* is a contact graph, understood as a system that individuates items against its own rest. The *cell an item registers* in a consumer is the item’s resting cut computed in that consumer’s graph.

Theorem 8.6 (Consumer-relativity: distinct consumers register distinct cells, each correct). *The same item registers different cells in consumers with different graphs, and each registration is a correct individuation at that consumer’s floor. No registration is privileged as “the” identity, and there is no point-valued identity of which the registrations are approximations.*

Proof. By Theorem 8.5 the registered cell is the item’s resting cut in the consumer’s graph; different graphs yield different cuts—a sparse graph individuates the item against few contrasts, giving a thin cell; a dense one against many, giving a rich cell. Each is the minimum-cut individuation in its own graph and hence correct there, attaining that graph’s floor by Theorem 3.4. Were there a privileged point-valued identity, a registration differing from it would be incorrect; but each registration is correct in its own graph, and by Theorem 8.1 no point-valued identity exists to be approximated. Hence none is privileged. $\square$

Theorem 8.7 (No runtime verdict). *A system whose vocabulary over its graph consists of identifying an item, reading its committed values, transforming internally, and committing new values, cannot issue a verdict of success or failure on a determination.*

Proof. A verdict is a comparison of an achieved state to an expected one. The stated vocabulary contains no term for an expectation: the

graph stores what was committed (Theorem 6.1), not what should have been committed, and by Theorem 8.6 there is no privileged registration against which an achieved cell could be measured. The system therefore holds no expected state and possesses no comparison operator, so there is no quantity it could compute having the meaning of a verdict. Adjudication is possible only in a consumer holding an expectation of its own, and such a judgement is that consumer's, not the system's. $\square$

Remark 8.8 (Why this is a feature of the design). Theorem 8.7 is the formal content of a runtime that runs to completion and records anomalies rather than raising them. An unexpected determination is committed as a value like any other and is available to any consumer that reads it. What is refused is not error handling but the runtime's arrogation of a judgement that Theorem 8.6 shows belongs to consumers. Section 14 supplies the honest terminal states such a system *can* report.

9 The identity criterion

We now assemble Section 5 and Section 6 into a criterion, and test it against the classical puzzles of persistence through change (Lewis, 1976; Plutarch, 75; Wiggins, 2001).

9.1 Neither component suffices alone

Proposition 9.1 (χ alone is not an identity criterion). *There exist contact graphs Γ and Γ' with $\chi(\Gamma) = \chi(\Gamma')$ that are not the same individual under any reasonable criterion: take Γ' a faithful reconstruction of Γ built independently, or a copy of Γ authored alongside it.*

Proof. χ is a weighted-graph invariant (Theorem 5.5), so any Γ' isomorphic to Γ has $\chi(\Gamma') = \chi(\Gamma)$. Isomorphic graphs include reconstructions and copies. Since a copy is uncontroversially a distinct individual from its original—they may be operated on independently, and a change to one does not change the other— χ does not discriminate individuals. $\square$

Proposition 9.2 (M alone is not an identity criterion). *The record is a non-negative integer carrying no information about what is being counted; two unrelated structures may stand at equal records.*

Proof. Immediate from Theorem 6.1: M counts committing acts and is determined by their number alone. $\square$

Definition 9.3 (Individual). An *individual* is the pair (χ, M) carried by a structure: its conserved invariant together with its committed record. Two structures are *the same individual* at two moments if and only if there is a chain of committing acts joining them along which χ is preserved and M is strictly increasing, with no intervening stage at which any unit of the structure fails to be terminated in the sense of Theorem 9.4.

9.2 Termination is the operative criterion

Definition 9.4 (Terminated and open units). A *unit* is a connected item set U with its induced contact structure. It is *terminated* if its resting cut against the medium is attained, giving a definite separation cost $\sigma(U) \geq \beta$; it is *open* if no such cut is attained.

Theorem 9.5 (Only terminated units are individuated). *A unit has a defined separation cost, hence a defined identity in the sense of Theorem 5.3, only if terminated. An open unit has no attained cut against the medium and therefore no defined identity.*

Proof. Identity is the resting cut (Theorem 5.3), which by Theorem 2.4 is the attained minimiser of the separation cost. If no such cut is attained, the quantity is undefined and no invariant is assigned. $\square$

Remark 9.6 (Termination is not free). Committing a completion boundary—making a collection of parts into a definite unit—is itself a cut and deposits residue $\geq \beta$ (Theorem 3.5). The act of making a thing a definite thing already pays the floor. This is what makes Theorem 9.7 below non-trivial.

Theorem 9.7 (Continuity of termination, not of parts). *Let a structure undergo a sequence of committing acts.*

- (i) Gradual replacement. *If parts are replaced one at a time in such a way that the unit is terminated at every intermediate stage, then χ is preserved, M increases continuously, and the individual persists—even if every part is eventually replaced.*

Table 1: The identity criterion adjudicating the classical cases. χ is preserved in all three; the record and the termination condition separate them.

Case	χ	record	termination condition
parts replaced singly	preserved	continuous chain	yes
disassembled, rebuilt	preserved	chain via open stage	no
copy authored alongside	identical	separate lineage	no

- (ii) Disassembly and reconstruction. *If the structure is fully disassembled and then reassembled to identical specification, the resulting structure has χ equal to the original but is a distinct individual.*
- (iii) Copying. *A structure authored alongside an original, to identical specification, is a distinct individual regardless of χ .*

Proof. (i) Each replacement is a committing act on a terminated unit; by Theorem 5.2 the invariant is preserved under any re-encoding preserving the cut structure, and replacing a part while preserving the contacts and weights is such a re-encoding. By Theorem 6.4 the record strictly increases at each act, and by hypothesis no stage leaves the unit open, so Theorem 9.3 is satisfied along the chain. That the parts are eventually wholly replaced is irrelevant: by Theorem 5.3 the identity is borne by the cut structure, not by any vertex.

(ii) Disassembly leaves the unit open (Theorem 9.4): no resting cut is attained, so by Theorem 9.5 no invariant is assigned at that stage. Theorem 9.3 requires that no intervening stage be open, which fails. Equality of χ after reassembly follows from Theorem 5.2, since the reassembled structure is isomorphic to the original; but reassembly is a fresh termination, paying the floor again (Theorem 9.6) and committing new cuts at strictly higher record. By Theorem 6.4 the record cannot be restored to its pre-disassembly value, so the reassembled structure stands at a record whose chain to the original passes through an open stage. Hence it is a distinct individual.

(iii) A copy is isomorphic, so χ agrees by Theorem 5.2. But no chain of committing acts joins the two: the copy’s record was advanced by its own authoring acts, not by acts on the original, and by Theorem 6.4 neither record is a continuation of the other. Theorem 9.3 fails. $\square$

Corollary 9.8 (Snapshot and restore does not preserve an individual). *Serialising a structure and later reconstructing it from the serialisation produces a structure with the same χ and a distinct individual.*

Proof. Serialisation followed by reconstruction is disassembly followed by reassembly to specification, which is Theorem 9.7(ii): the serialised form is not a terminated unit in the graph—no resting cut of the structure is attained while it is a byte string—so the chain passes through an open stage. $\square$

Corollary 9.9 (Accretion is the only identity-preserving update). *An update that adds items, contacts, and committed values to a structure while leaving every unit terminated preserves the individual. An update that tears down and rebuilds does not, however faithful the rebuild.*

Proof. The first is Theorem 9.7(i); the second is Theorem 9.7(ii). $\square$

Remark 9.10 (Engineering consequences). Theorems 9.8 and 9.9 are the two practical commitments the criterion imposes on any system built on this account. Continuity across restarts must be obtained by persisting the *record* and accreting onto it, never by reloading a serialised state as though it were the same structure. Schema and content migrations must be performed accretively; a rebuild is a replacement, not a refactor. Both are checkable predicates, not matters of taste: the first asks whether the record is continuous, the second whether any unit was left open.

Part II

The Causal Propagation Table

10 Alignment, accountability, and propagation

Part I fixed items at rest. We now make determination dynamic and construct the object whose two evaluation regimes are the subject of the paper.

10.1 Alignment

Definition 10.1 (Alignment; alignment score). Fix a contact graph and a target item x^* . The

alignment of an item x to x^* is the minimum-cut weight separating them,

$$\sigma(x, x^*) = \min_{S \ni x, x^* \notin S} w(\partial(S)) \in [\beta, \Omega], \quad \Omega := w(E)$$

and the *alignment score* is $a(x, x^*) := \sigma(x, x^*)/\Omega \in (0, 1]$.

Lemma 10.2 (The floor bounds alignment below). *For distinct items $x \neq x^*$, $a(x, x^*) \geq \beta/\Omega > 0$: no two distinct items are perfectly aligned.*

Proof. Immediate from Theorem 3.4: any x – x^* cut has weight $\geq \beta$, and cuts are non-empty by connectivity. $\square$

Remark 10.3 (Uncertainty without an imported calculus). The quantity ordinarily carried by a probability, a fuzzy membership, or a posterior—a number in $[0, 1]$ recording how sure—is carried throughout this paper by $a(\cdot, \cdot)$ and its distance to the floor β/Ω . This quantity is derived (an exact minimum-cut weight), bounded below by $\beta/\Omega > 0$, and parameter-free: no prior is set and no membership function is tuned. Where a standard calculus *posits* an uncertainty, the floor *forces* one. No theorem depends on the comparison.

10.2 The residual, accountability, and the floor of a system

Definition 10.4 (System floor). The *system floor* of a contact graph is $\beta^* := \min_{v \in V \setminus \{m\}} \sigma(v)$, the least separation cost of any item against the medium.

Definition 10.5 (Accountability). Fix a tolerance $\varepsilon \geq 0$. A determination of x^* from v_0 is ε -*accountable* when

$$\sigma(v_0, x^*) \leq \beta^* + \varepsilon\Omega. \quad (1)$$

The *accountable class* $\mathcal{A}(v_0, x^*, \Pi)$ is the set of propagations under Π (Theorem 10.7) from v_0 to x^* satisfying (1); it may be empty.

Remark 10.6 (Two independently movable quantities). The left side of (1) is local to the queried pair: a single minimum cut between v_0 and x^* . The right side, at $\varepsilon = 0$, is a minimum over *every* item in the system, including items lying on no path between the two. The whole of Section 15 is the observation that these two can be moved independently and that no stored assertion records either.

10.3 Processes and propagations

Definition 10.7 (Process graph; propagation).

A *process graph* is a contact graph $\Pi = (V, E, w; I, x^*)$ together with a set $I \subseteq V$ of *seeds* and a designated *target* $x^* \in V$. A *propagation* of a seed $v_0 \in I$ through Π is a walk $W = (v_0, e_1, v_1, \dots, e_k, v_k)$ along edges of Π with $v_k = x^*$; each edge e_i is a contact operation (Theorem 7.1). The *trivial* or *rest* process $\Pi_{\text{rest}}(v)$ has $I = \{v\}$, $x^* = v$, and the single available operation $\{v, m\}$.

Theorem 10.8 (Convergence admissibility). *A propagation is admissible if and only if it is convergent: its terminal item is x^* . Interior items are entirely unconstrained—an interior item may have arbitrarily large alignment to the target, i.e. may be maximally unlike it, without rendering the propagation inadmissible. Inadmissibility arises exclusively from failure to reach x^* , never from local misalignment along the way.*

Proof. By Theorem 10.7 a propagation terminates at x^* by construction, and its terminal alignment is $a(x^*, x^*)$, the least value permitted by Theorem 10.2; so every propagation in the defined sense is convergent. A walk failing to reach x^* has terminal item $v_k \neq x^*$ and is not a propagation to x^* ; it is inadmissible as a determination of x^* . Nothing in Theorem 10.7 constrains the alignment of any interior item: the walk is defined purely by adjacency and by its terminus. Hence admissibility depends exclusively on the terminal condition. $\square$

Theorem 10.9 (Path opacity). *Let W_1, W_2 be two propagations of the same seed to the same target, differing in their interior items. Then no invariant computed from the seed and target alone—the seed, the target, the terminal alignment, or the target’s resting cut—distinguishes W_1 from W_2 . The interior of a propagation is unobservable from its result.*

Proof. Every listed invariant is a function of the endpoints alone, hence identical on the two propagations by hypothesis. The interior data are, by Theorem 10.8, unconstrained given the endpoints, so they are not recoverable as a function of the endpoints; an invariant computed only from the endpoints therefore cannot distinguish them. $\square$

Corollary 10.10 (The deliverable is a class, not a path). *For fixed seed, target, and process, the admissible propagations form an equivalence class*

under “shares seed and target, and converges,” every member of which is indistinguishable from every other by any test of the result alone. A determination’s deliverable is properly this class.

Proof. Immediate from Theorems 10.8 and 10.9. $\square$

Remark 10.11 (Path opacity is not a limitation). Theorem 10.9 says the interior is not recoverable from the result. It does not say the interior is lost: the committed edges of W_1 and W_2 differ in the graph, so the two propagations leave different structures behind even though their outputs agree. This is the distinction that Section 7.2 turns on. History is retained as committed structure, and is precisely not retained as material available for re-representation—which is why Theorem 7.11 does not apply to a system built this way.

10.4 Representation mobility

Definition 10.12 (Representation; representation fibre). Fix an item v and a target x^* , and write $a = a(v, x^*)$. A *representation* of v at dimension N is a tuple $(s_1, \dots, s_N) \in \mathbb{R}^N$ satisfying the *averaging constraint* $\frac{1}{N} \sum_j s_j = a$, with individual components otherwise unconstrained in $\mathbb{R}$ —in particular, a component may lie outside $(0, 1]$. The *representation fibre* of v at dimension N is the set of all such tuples.

Theorem 10.13 (Representation mobility). *For every item v and every $N \geq 2$:*

- (i) *the representation fibre is non-empty and infinite;*
- (ii) *switching from one representation to another commits no new contact and leaves the record unchanged;*
- (iii) *any computation depending on v only through $a(v, x^*)$ is invariant under representation switching.*

Proof. (i) The averaging constraint is a single linear equation in N unknowns; for $N \geq 2$ its solution set is an affine hyperplane of $\mathbb{R}^N$, hence non-empty and infinite, and any $(s_1, \dots, s_{N-1})$ extends uniquely by $s_N = Na - \sum_{j < N} s_j$. (ii) A switch replaces one tuple in the fibre with another; both denote the same item at the same alignment, so the switch introduces no new item, no new contact, and hence no committing act. (iii) By hypothesis the computation is a function of a alone, and every fibre member averages to the same value. $\square$

Remark 10.14 (Components may be individually inadmissible). Theorem 10.12 deliberately permits components outside the range that any single admissible item could occupy: only the *mean* is constrained. This is not a loophole but the substance of the definition. It permits intermediate decompositions that are individually impossible while their aggregate is exactly the required alignment—and by Theorem 10.13(iii) nothing downstream can tell the difference, since downstream computations see only the mean. Theorem 10.8 is the dynamic counterpart: interior steps of a propagation are unconstrained provided the walk converges.

10.5 The table

Definition 10.15 (Causal propagation table). The *causal propagation table* of a contact graph is the map

$$\mathcal{T}(v_0, x^*, \Pi) := \begin{cases} \partial(S^*(x^*)) & \text{if } \mathcal{A}(v_0, x^*, \Pi) \neq \emptyset, \\ \emptyset & \text{otherwise,} \end{cases}$$

returning, for a seed, a target, and a process, the accountable terminal cut of the propagation—or the verdict that no propagation from v_0 to x^* under Π is accountable.

Proposition 10.16 (The table’s outputs are cuts of weight $\geq \beta$). *Every non-empty output of $\mathcal{T}$ is a cut of weight $\geq \beta$ in the sense of Theorem 2.4, hence an object of the same type as the resting cut of an item.*

Proof. By Theorem 10.15 a non-empty output is the resting cut of x^* , which by Theorem 3.4 has weight $\geq \beta$. $\square$

Proposition 10.17 (Edges are products of a run). *The causal relation on items is not a fixed attribute of the graph. A causal edge $u \rightarrow v$ obtains in a propagation exactly when, in that propagation, a value committed at u was read at v . Consequently the causal edge set is a product of a propagation, not an input to it, and two propagations over the same item set may induce different causal edge sets.*

Proof. By Theorem 6.1 an edge becomes committed only by a cut event, and by Theorem 10.7 which cut events occur is determined by the walk taken. Two walks with different interiors—admissible by Theorem 10.8—commit different edges. $\square$

Part III

The Directional Pair

11 The catalogue is the table at rest

We now prove the paper’s central theorem. It says that a catalogue of items at rest and a machinery of propagation under process are not two objects but one object at two settings of a single argument.

11.1 The resting catalogue

Definition 11.1 (Resting catalogue). The *resting catalogue* of a contact graph is the assignment, to each item v , of its resting cut against the medium,

$$\text{cell}(v) := \partial(S^*(v)), \quad \sigma(v) \geq \beta.$$

A catalogue *cell* is the name of this cut: the invariant region by which v is told from the rest, at rest (Theorem 5.3).

Theorem 11.2 (A catalogue cell is an accountable output). *For each item v , the cell $\text{cell}(v)$ equals the output of the table run on v against the medium under the trivial process:*

$$\text{cell}(v) = \mathcal{T}(v, v, \Pi_{\text{rest}}(v)).$$

Equivalently, a catalogue entry is the convergent terminal cut of the resting propagation.

Proof. Take $\Pi_{\text{rest}}(v)$ as in Theorem 10.7: seeds $I = \{v\}$, target $x^* = v$, and the single available contact $\{v, m\}$. The propagation is the one cut event individuating v from m ; it is admissible by Theorem 10.8 since it terminates at its target, and its terminal accountable object is by Theorem 10.15 the resting cut $\partial(S^*(v))$. This is precisely $\text{cell}(v)$ by Theorem 11.1. Accountability at the target is immediate: taking $v_0 = x^* = v$ gives $\sigma(v, v)$ realised by the minimum cut against the medium, and by Theorem 10.4 this is $\geq \beta^*$ with equality attained at the item realising the system floor, so (1) is satisfiable at $\varepsilon = 0$ and satisfied for all $\varepsilon \geq 0$ at the realising item. $\square$

Theorem 11.3 (Directional identity: the catalogue is the table at rest; the table is the catalogue propagated). *The resting catalogue is the restriction of the causal propagation table to trivial processes:*

$$\text{cell}(v) = \mathcal{T}(v, v, \Pi_{\text{rest}}(v)) \quad \text{for every item } v.$$

Conversely, for a nontrivial process Π the output $\mathcal{T}(v_0, x^, \Pi)$ is an object of the same type as a catalogue cell—an accountable cut of weight $\geq \beta$, the convergent terminal individuation of the tracked item—computed under propagation rather than at rest.*

Proof. The forward direction is Theorem 11.2. For the converse: by Theorem 10.8 every admissible propagation terminates at x^* , whose individuating datum is the resting cut $\partial(S^*(x^*))$ of weight $\geq \beta$ by Theorem 3.4; by Theorem 10.16 this is the same type of object as $\text{cell}(\cdot)$ in Theorem 11.1. The two differ only in that the cut is the terminus of a nontrivial propagation rather than of the rest contact. Hence the table’s output for a process is a catalogue-grade cut computed under propagation. $\square$

Corollary 11.4 (One object, two directions). *A catalogue and a propagation machinery are not two systems requiring an interface. They are one map $\mathcal{T}$ evaluated at two settings of its process argument, whose outputs inhabit a single type. No translation between them is required or possible: there is nothing to translate.*

Proof. Immediate from Theorem 11.3: both regimes are values of the same map, and by Theorem 10.16 both outputs are cuts of weight $\geq \beta$. $\square$

Remark 11.5 (The canonical instance). The familiar catalogue of chemical elements (Mendeleev, 1869; Scerri, 2007) is, on this reading, a resting catalogue in the sense of Theorem 11.1: each entry is the name of an item’s minimum-sufficient individuation against everything else, established by an array of resting determinations. The causal propagation table extends it to arbitrary processes, and by Theorem 11.3 the output of a process is an entry of the same kind, computed for that process. The catalogue’s correctness is inherited, not contested; the table adds only the propagated cells the catalogue does not list, because they are process-dependent. No theorem depends on the example.

11.2 The pair

We now name the two directions and prove the properties of their pairing.

Definition 11.6 (Catalogue side; process side; the directional pair). Let Γ be a contact graph and

$\mathcal{T}$ its causal propagation table. The *catalogue side* G is $\mathcal{T}$ restricted to trivial processes; the *process side* L is $\mathcal{T}$ evaluated at nontrivial processes. The *directional pair* is the ordered pair $P = (G, L)$ together with the *deposit map* that commits the residue of each process-side propagation into Γ as new contacts (Theorem 6.1).

Theorem 11.7 (The pair is one individual). *The directional pair $P = (G, L)$ carries a single invariant and a single record: $\chi(P) = \chi(\Gamma)$ and $M(P) = M(\Gamma)$, where Γ is the shared contact graph. Process-side propagation advances $M(P)$ and preserves χ -conservation in the sense of Theorem 5.2 under every relabelling.*

Proof. By Theorem 11.6 both sides are evaluations of one map over one graph, so there is one graph to carry an invariant and a record. That $\chi(\Gamma)$ is a weighted-graph invariant is Theorem 5.5. That process-side propagation advances the record is Theorem 6.1 together with Theorem 10.7: each contact operation of a propagation is a cut event. Conservation under relabelling is Theorem 5.2, which is a statement about the graph and is unaffected by which evaluation regime produced a given committed edge. $\square$

Theorem 11.8 (The pair blunts where neither half alone can). *Let L perform a propagation under a nontrivial process. Then the residue of every contact operation of that propagation is deposited into Γ , so that:*

- (i) $M(P)$ strictly increases;
- (ii) the uncommitted capacity of the items involved strictly decreases;
- (iii) consequently Theorem 7.5 is satisfied by the pair, and Theorem 7.6 is not violated.

Proof. By Theorem 10.7 a propagation is a sequence of contact operations, each a cut event committing an edge of Γ . By Theorem 7.2 each such operation deposits its residue into both endpoints, and by Theorem 11.6 the deposit map commits these into Γ . (i) is Theorem 6.1; (ii) is Theorem 7.3 together with the removal of the committed edge from the uncommitted set; (iii) is then Theorem 7.5 applied with Γ as the structure that carries the instrument's capacity and record. Since the pair's record advances and its capacity is consumed, the pair does not assert either property of Theorem 7.6. $\square$

Corollary 11.9 (The catalogue side is the process side's identity). *The process side L , considered alone, has no invariant and no record: by Theorem 11.6 it is a family of evaluations, not a structure. Paired with G , it acquires both, because the graph is where its propagations deposit. Hence (χ, M) of the pair is carried entirely by the catalogue side, and it is in this precise sense that the catalogue is the process side's identity.*

Proof. L is by Theorem 11.6 the map $\mathcal{T}$ evaluated at nontrivial processes; a map is not a contact graph and carries no $\sigma(\cdot)$ of its own, hence no χ by Theorem 5.4 and no committed record by Theorem 6.1. Theorem 11.7 assigns the pair the graph's invariant and record, and Theorem 11.8 shows process-side activity advances them. The identification follows. $\square$

Remark 11.10 (Why this is not integration). The engineering instinct on encountering a stateless determiner and a structured store is to interface them: define a protocol, translate between representations, decide which is authoritative. Theorem 11.4 shows this misdescribes the situation. There is one map. The store is what it returns at rest and the determiner is what it returns under process, and by Theorem 10.16 the two returns are the same type of object. What connects them is not a protocol but the deposit map of Theorem 11.6, which is not a translation but the ordinary commitment of a cut (Theorem 6.1). Nothing crosses a boundary because there is no boundary to cross.

Remark 11.11 (Reading the direction). The word *directional* names the asymmetry in Theorem 11.3. The two regimes are not symmetric alternatives: the resting regime produces the fixed points of individuation, and the process regime produces cells the resting regime cannot, precisely because they depend on a process. The flow of residue is likewise directional—from propagation into the catalogue, by Theorem 11.8—and never the reverse, since by Theorem 6.4 committed structure is not un-committed. The catalogue accretes; the process side consumes and deposits.

11.3 What the pair inherits

We record, as corollaries, the properties the pair inherits from Parts I and II, since these are what a builder must honour.

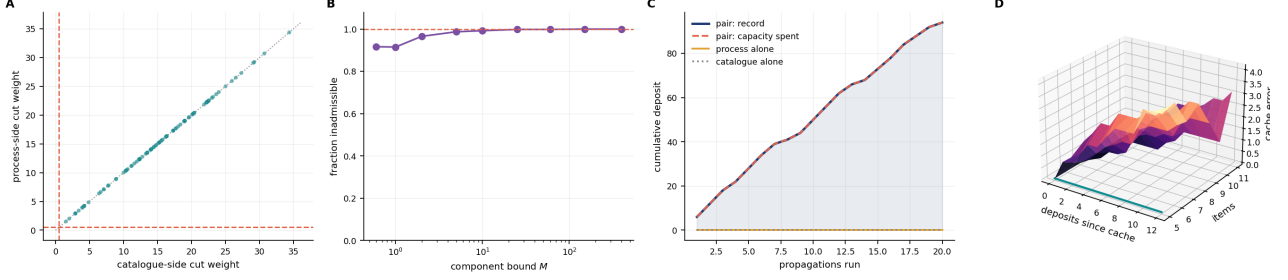


Figure 3: One Table, Two Directions; and the Deposit That Makes the Pair an Individual. (A) Process-side cut weight on the linear y -axis versus catalogue-side cut weight on the linear x -axis, for 200 outputs drawn from 50 graphs, with the floor marked on both axes (crimson dashed) and the identity diagonal in grey. Every output of both regimes is a cut of weight $\geq \beta$: 0 type violations. The two regimes do not merely interoperate—they return objects of one type, which is the converse half of Theorem 11.3 (Directional Identity). The forward half is verified separately on all 320 items of the same graphs, with 0 mismatches. (B) Fraction of admissible representations carrying at least one component outside the range a single admissible item could occupy, on the linear y -axis versus the component bound M on a log x -axis (0.6 to 400), 3000 draws per point. The fraction climbs to and remains at 1.0 (crimson reference) once M exceeds a few units: inadmissible components are the generic case, not an exception, exactly as Remark 10.14 asserts. Over 200 independent trials the mean is preserved to a maximum error of 5.0×10^{-15} while 98.5% of representations carry an inadmissible component. (C) Cumulative deposit on the linear y -axis versus propagations run on the linear x -axis. The pair’s record (navy) and consumed capacity (crimson dashed) rise together across the run; the process alone (amber) and the catalogue alone (grey dotted) remain flat at zero. The shaded band between them is what Theorem 11.8 supplies and neither half supplies on its own; the pattern held in all 40 trials. (D) Error of a cached determination as a three-dimensional surface (magma) over deposits since caching on the x -axis (0 to 12) and item count on the y -axis (5 to 11), elevation 24° azimuth -56° . The surface rises away from the teal zero-line, which is the pair recomputing against the current graph. Across 40 trials the cache went stale in 21 and the pair in 0: the geometry of Proposition 7.10 against Corollary 11.13.

Corollary 11.12 (No re-presentation). *In a directional pair, a prior propagation is retained as committed edges of Γ and is never available as material for a later propagation to reason about. Hence Theorem 7.11 does not apply: there is no operation that converts a prior propagation into content.*

Proof. By Theorem 11.6 the deposit map commits residue as contacts. By Theorem 7.4 a committed edge is not re-committed: a second commitment is either a no-op or a distinct cut event at higher record. There is therefore no operation returning a prior propagation as an input, and the hypothesis of Theorem 7.11—that P is re-presented as content—is unsatisfiable in the pair. $\square$

Corollary 11.13 (No staleness). *Every determination the pair returns is computed against Γ as it stands at the record of the determination.*

Proof. By Theorem 10.15 a table output is a minimum cut computed in the current graph, and by Theorem 11.8 every prior propagation has already been committed into that graph. Hence no output is a determination of an earlier configuration, and the hypothesis of Theorem 7.10 is unsatisfiable. $\square$

Corollary 11.14 (Repeated queries need not agree, and this is correct). *Two determinations of the same target at two records need not coincide. The divergence is not error: by Theorem 11.8 the first determination committed cuts that altered Γ , so the second is computed against a different graph.*

Proof. Immediate from Theorem 11.13 and Theorem 6.4: the two determinations are computed at distinct states, and Theorem 10.15 makes the output a function of the graph. $\square$

Remark 11.15 (Reproducibility relocates). Theorem 11.14 abandons a familiar expectation deliberately. If repeated determinations need not agree, what is reproducible? The answer is the *protocol and its provenance*: which items were present, which process was run, which contacts were available. Another party reproduces a determination by reconstituting the same process against the same catalogue and expects, not the same cut, but a determination of the same protocol whose differing cuts are themselves data (Moreau and Groth, 2013; Peng, 2011; Sandve

et al., 2013). This is the sense in which an experimental result is reproducible: not that the readings coincide, but that the protocol can be re-enacted and the readings compared (Open Science Collaboration, 2015). No theorem depends on the reading.

Part IV

Construction and Probing

12 Construction: the term map is the only declared input

Everything in Parts I–III is a theorem about a contact graph. Nothing so far says where the graph comes from. We now isolate that question, show it is a single well-defined input, and prove that the calculus is correct for any value of it—which is what makes a fallible constructor safe to use.

12.1 The term map

Definition 12.1 (Term map; induced contact graph). Let $\mathcal{U}$ be a finite set of *sources*—documents, records, measurements, prior determinations. A *term map* is a function $\tau: \mathcal{U} \rightarrow 2^{\mathcal{D}}$ assigning to each source the finite set of *distinctions* it draws, valued in a distinction alphabet $\mathcal{D}$. The *induced contact graph* Γ_τ has vertex set $\mathcal{U} \cup \{\mathbf{m}\}$, a contact $\{u, v\}$ whenever $\tau(u) \cap \tau(v) \neq \emptyset$ with weight $w(\{u, v\}) = f(|\tau(u) \cap \tau(v)|)$ for a fixed strictly increasing $f: \mathbb{N}_{>0} \rightarrow \mathbb{R}_{\geq \beta}$, and $\mathbf{m}$ adjacent to every source at weight $\geq \beta$.

Remark 12.2 (What the term map is and is not). τ is a reading of unstructured material. It is not a judgement about truth, relevance, or importance: it says only which distinctions a source draws. It is the one place where a component external to the calculus enters, and Theorem 12.3 below bounds exactly what damage such a component can do.

Theorem 12.3 (The calculus is correct for any term map). *Let τ be any term map and Γ_τ its induced contact graph. Then every result of Parts I–III holds of Γ_τ : the floor is positive, the character invariant is conserved and non-local, the record is monotone, individuation is symmetric and self-blunting, the table is well-defined, and*

the directional identity holds. No result requires τ to be accurate, complete, or agreed.

Proof. By Theorem 12.1 Γ_τ is a finite weighted graph with a medium vertex adjacent to every item and all weights $\geq \beta$, i.e. a contact graph in the sense of Theorem 2.2. Premise 1 holds by finiteness of $\mathcal{U}$ and Premise 2 is a premise about the medium, unaffected by τ . Every theorem of Parts I–III is derived from Theorem 2.2 together with Premises 1 and 2 alone, and none inspects the provenance of an edge. Hence all hold. $\square$

Corollary 12.4 (Extraction error coarsens; it does not corrupt). *An inaccurate term map yields a contact graph whose cells are coarser or finer than they would otherwise be, and whose determinations are correspondingly coarser or finer. It cannot make a determination unsound relative to the graph it induces, because soundness is a property of the graph and Theorem 12.3 holds for every graph.*

Proof. Immediate from Theorem 12.3: the conclusions are theorems about Γ_τ , whatever τ was. $\square$

12.2 The floor as a quality readout

Theorem 12.4 says an error in τ is visible as coarseness. We now show the coarseness is measurable, and measurable *without a reference graph to compare against*, which is the practical value of the result.

Definition 12.5 (Refinement of term maps). A term map τ' *refines* τ , written $\tau' \preceq \tau$, if for all u, v , $\tau'(u) \cap \tau'(v) \supseteq \tau(u) \cap \tau(v)$: every distinction τ draws in common between two sources, τ' also draws.

Theorem 12.6 (Floor monotonicity). *If $\tau' \preceq \tau$ then $\beta^*(\Gamma_{\tau'}) \geq \beta^*(\Gamma_\tau)$: a finer term map induces a system floor at least as large. Equivalently, the realised system floor is monotone in the refinement order on term maps.*

Proof. Fix an item v . By Theorem 12.5 every contact present in Γ_τ is present in $\Gamma_{\tau'}$ with weight at least as large, since f is strictly increasing and the shared-distinction count does not decrease. Any cut separating v from m in $\Gamma_{\tau'}$ therefore has weight at least that of the corresponding cut in Γ_τ , so the minimum over cuts satisfies $\sigma_{\Gamma_{\tau'}}(v) \geq \sigma_{\Gamma_\tau}(v)$. Taking the minimum over items (Theorem 10.4) preserves the inequality. $\square$

Corollary 12.7 (Ground-truth-free quality signal). *The realised system floor $\beta^*(\Gamma_\tau)$ is computable from Γ_τ alone by $|V|$ minimum-cut computations (Ford and Fulkerson, 1956; Stoer and Wagner, 1997), and by Theorem 12.6 it is monotone in the quality of τ under refinement. It therefore serves as a quality readout for the constructing component that requires no reference graph and no labelled data.*

Proof. Computability is Theorem 2.4 together with the exactness of maximum flow; monotonicity is Theorem 12.6. $\square$

Remark 12.8 (The shape of the guarantee). Theorem 12.7 is a monotonicity statement, not a calibration. It says that refining the term map cannot lower the realised floor, so a system whose floor is not rising under attempted refinement is not being refined. It does not say that a higher floor is always better—an over-connected graph in which everything shares distinctions with everything has a high floor and poor discrimination—so the readout is a signal to be read alongside cell sizes, not a scalar to be maximised. We state this explicitly because the temptation to optimise it directly is obvious and would be an error.

12.3 Construction is itself a determination

Construction is not outside the calculus. It is a determination against the medium, and the results of Sections 13 and 14 govern it.

Principle 12.9 (Construction discipline). A term map should be produced by no fewer than three constructors of distinct kind, run to closure in the sense of Theorem 14.2, with disagreement between constructors recorded as contested closure (Theorem 14.5) rather than resolved by adjudication—which by Theorem 8.7 the system cannot perform anyway.

Remark 12.10 (Why three, and why distinct). The count and the diversity requirement are not stylistic. Theorem 13.11 proves that a support structure of fewer than three members is not robust to the failure of a single member, and Theorem 13.8 proves that repeated application of one constructor accrues resolution strictly more slowly than application of distinct ones and may fail to saturate at all. Two constructors agreeing is mutual definition without external check; five prompts to one constructor is a single constructor.

Remark 12.11 (Carrying structure, not material). Once Γ_τ is built, the sources $\mathcal{U}$ need not be carried. By Theorem 6.6 the committed contacts retain the *links*—the information reading—whereas re-supplying the source material would present the deposits without their links, i.e. the entropy reading. Sources are regenerable from an external substrate given the distinctions that were drawn from them; committed structure is not regenerable from the sources, because by Theorem 12.1 the same source contributes different contacts in different graphs. Carry the structure; regenerate the material.

13 Probing: many determiners over one graph

A determination need not be made by a single mechanism. We develop the algebra of multiple determiners—*probes*—acting on one contact graph, and prove four things about them: how they compose, how many are required, how a finite budget should be divided among them, and why finding and pruning cannot be one operation.

13.1 Probes and probing power

Definition 13.1 (Probe). A *probe* toward a target x^* is a partial map $\gamma: V \rightharpoonup V$ on items such that, where defined, applying γ to an item x commits a contact and yields $\gamma(x)$; that is, a probe realises a contact operation (Theorem 7.1).

Definition 13.2 (Probing power). The *probing power* of γ toward x^* at an item x is the fraction of above-floor alignment it closes:

$$\kappa(\gamma, x, x^*) = \frac{a(x, x^*) - a(\gamma(x), x^*)}{a(x, x^*) - \beta/\Omega} \in [0, 1].$$

A probe is *inert* if $\kappa = 0$ and *absolute* if $\kappa = 1$.

Lemma 13.3 (Power is bounded). $\kappa \in [0, 1]$ for every probe, item, and target.

Proof. By Theorem 10.2, $a(\gamma(x), x^*) \geq \beta/\Omega$, so the numerator never exceeds the denominator. Non-negativity holds for probes directed toward x^* ; a probe that increases alignment is assigned $\kappa = 0$ by convention, not being a probe toward x^* in the operative sense. $\square$

Remark 13.4 (Probes are structurally unconstrained). Theorems 13.1 and 13.2 inspect nothing about a probe beyond its action on items and

the alignment it closes. A probe may be a local computation, a remote retrieval, a deterministic rule, or an inference by a trained model: none of the results below distinguishes these cases, because none of the definitions refers to anything but γ 's action. A classification of probes by kind is useful for scheduling—expected cost and latency differ—but carries no weight in the theory.

13.2 Composition is multiplicative

Theorem 13.5 (Multiplicative composition). For probes $\gamma_1, \dots, \gamma_n$ applied in sequence, each to the output of the last, with powers $\kappa_1, \dots, \kappa_n$ toward a fixed target, the composite power is

$$\kappa(\gamma_1 \diamond \dots \diamond \gamma_n) = 1 - \prod_{i=1}^n (1 - \kappa_i).$$

Proof. Let $r_0 = a(x_0, x^*) - \beta/\Omega$ be the initial above-floor gap. Applying γ_1 leaves gap $r_1 = (1 - \kappa_1)r_0$ by Theorem 13.2; applying γ_2 leaves $r_2 = (1 - \kappa_2)(1 - \kappa_1)r_0$; by induction $r_n = \prod_{i=1}^n (1 - \kappa_i)r_0$. The fraction of the initial gap closed is $1 - r_n/r_0$, which is the composite power by Theorem 13.2 applied to the chain as a single probe. $\square$

Corollary 13.6 (Diminishing returns of repetition). Repeating one probe of power κ a total of n times gives composite power $1 - (1 - \kappa)^n$: strictly increasing, convergent to 1, never reaching it for finite n . Each repetition closes a strictly smaller absolute fraction of the remaining gap than the last.

Corollary 13.7 (Saturation dichotomy). A chain of probes with powers $(\kappa_i)_{i \geq 1}$ drives the residual gap to zero in the limit if and only if $\sum_i \kappa_i = \infty$.

Proof. The residual fraction after n probes is $\prod_{i \leq n} (1 - \kappa_i)$. Taking logarithms, $\sum_{i \leq n} \log(1 - \kappa_i) \rightarrow -\infty$ if and only if $\sum_i \kappa_i = \infty$, since $-\log(1 - \kappa) \asymp \kappa$ for small κ and the comparison test applies; this is a dichotomy of Borel–Cantelli type (Borel, 1909; Cantelli, 1917). $\square$

Corollary 13.8 (Diversify, do not repeat). A chain of n distinct probes with powers $\kappa_1, \dots, \kappa_n$ attains composite power $1 - \prod_i (1 - \kappa_i)$, which strictly exceeds the composite power $1 - (1 - \kappa_j)^n$ of n repetitions of any single member γ_j whenever some $\kappa_i > \kappa_j$. A chain of near-duplicate probes accrues power slowly and, if its powers decay fast enough, fails to saturate even in principle (Theorem 13.7).

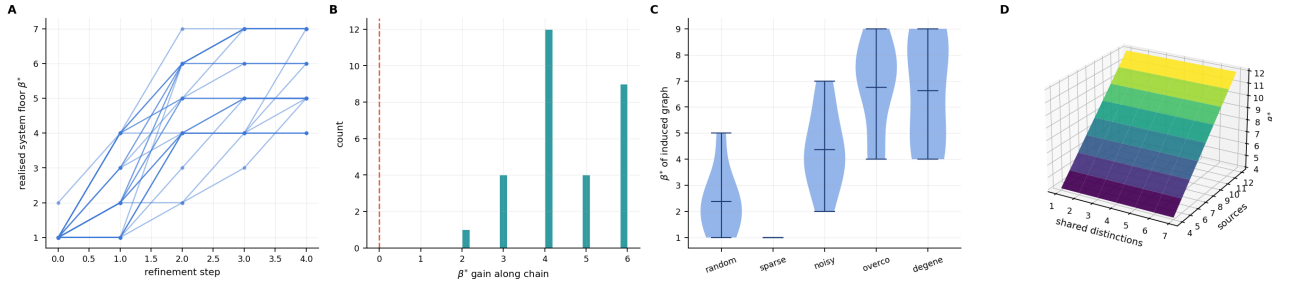


Figure 4: The Calculus Is Indifferent to the Quality of the Term Map, and the Realised Floor Reports That Quality Without a Reference. (A) Realised system floor β^* on the linear y -axis versus refinement step on the linear x -axis (0 to 4), for 26 of the 40 chains of term maps ordered by the refinement relation of Definition 12.5. Every chain is monotone non-decreasing; across all 40 valid chains there are 0 monotonicity violations. This is the empirical content of Theorem 12.6 (Floor Monotonicity). (B) Distribution of the floor gain from the coarsest to the finest map of each chain, 22 bins, with the crimson reference at zero. The support lies entirely to the right of it: no chain loses floor under refinement, and the gains cluster at 4 to 6 units. (C) Distribution of the realised floor by kind of term map, as violins over five deliberately adversarial families—random, sparse, noisy, over-connected, and degenerate (every source drawing identical distinctions), 60 maps in total. Every family induces a graph on which the floor, the edge bound, relabelling invariance, and cut non-emptiness all hold: 0 failures, the content of Theorem 12.3. The families differ markedly in the floor they induce, which is the coarsening of Corollary 12.4 made visible. Note that the degenerate family induces among the *highest* floors while discriminating worst: the floor is a monotone signal of refinement, not a quantity to maximise (Remark 12.8). (D) Realised floor as a three-dimensional surface (viridis) over the number of shared distinctions on the x -axis (1 to 7) and the number of sources on the y -axis (4 to 12), each cell the mean of five induced graphs, elevation 23° azimuth -60° . The surface is smooth and monotone in the shared-distinction axis: more shared structure yields a higher floor and correspondingly coarser cells, across the whole two-dimensional parameter region.

Proof. Both composite powers are given by Theorem 13.5; the comparison is the elementary fact that $\prod_i (1 - \kappa_i) < (1 - \kappa_j)^n$ when some factor is strictly smaller than $(1 - \kappa_j)$. Failure to saturate is Theorem 13.7 applied to a summable power sequence. $\square$

13.3 Coherence requires three

Multiplicativity presumes probes directed at one target. We now show that robustness against the failure of a single probe requires a support structure of at least three members.

Definition 13.9 (Support; support graph). For probes γ_i, γ_j toward a common target, say γ_j supports γ_i at threshold $\theta \in (0, 1]$ if the presence of γ_j 's result does not reduce, and to degree $\geq \theta$ reinforces, the alignment shift toward the target induced by γ_i . The support graph G has an edge $j \rightarrow i$ whenever γ_j supports γ_i at θ .

Theorem 13.10 (Linear support fails). *No finite chain $\gamma_1 \rightarrow \gamma_2 \rightarrow \dots \rightarrow \gamma_n$ in which each probe's contribution is justified only by the next, and the terminal probe by nothing internal to the chain, is robust to removal of the terminal probe.*

Proof. The terminal probe γ_n has no incoming support edge in G ; its contribution rests on nothing internal to the chain. Removing γ_n leaves γ_{n-1} unsupported, and by induction each successive removal leaves the next probe unsupported. The chain's claim to be grounded is contingent on retaining every member, and a single removal at the terminus collapses the justification with no internal check to prevent it. $\square$

Theorem 13.11 (Coherence requires three mutually supporting probes).

- (i) Necessity. *If a set of probes brings a determination's terminal alignment within any margin of the floor and remains so after the failure of any single probe, its support graph contains a directed cycle of length ≥ 3 .*
- (ii) Sufficiency. *Three probes mutually supporting one another at threshold $\theta > \frac{1}{2}$, forming a strongly connected triangle in G , ground the determination robustly: their combined shift toward the target survives the removal of any single member.*

Proof. (i) By Theorem 13.10 no acyclic support structure is robust to removal of its terminal member; an acyclic G has a source probe with no incoming support, and the same argument applies

to it. A 1-cycle is self-support and vacuous. A 2-cycle is mutual definition between exactly two probes with no independent third check, and fails the majority condition $\theta > \frac{1}{2}$: removing either leaves one probe, itself unsupported. Hence the shortest support cycle compatible with robustness has length ≥ 3 .

(ii) In a strongly connected triangle each probe is supported by the other two; with $\theta > \frac{1}{2}$ the two supporters jointly exceed any single dissenting signal. Removing one leaves the remaining two still mutually supporting each other directly, above a reduced but positive effective threshold, so the shift toward the target persists. $\square$

Corollary 13.12 (Coherence is ordinally decidable). *Whether a set of probes is coherent in the sense of Theorem 13.11 is decidable from the signs of pairwise support relations alone—whether each pair reinforces or undercuts the other's shift—with no access to probing-power magnitudes.*

Proof. Theorem 13.11 is stated and proved entirely in terms of the directed-graph structure of G (cycle existence and length) and the threshold comparison $\theta > \frac{1}{2}$, both determined by the signed support relation of Theorem 13.9 and not by any numerical power. $\square$

13.4 Dividing a finite probing budget

Probes cost. We now solve the allocation problem, first in the discrete regime and then in the continuous one, and note precisely when each applies.

Definition 13.13 (Probe selection instance). Given probes $\gamma_1, \dots, \gamma_k$ with floors β_i (the residual each leaves) and per-invocation costs $c_i > 0$, and a budget $B > 0$, the *selection problem* is to choose $S \subseteq \{1, \dots, k\}$ minimising the composite residual $\Omega \prod_{i \in S} (1 - \beta_i/\Omega)$ subject to $\sum_{i \in S} c_i \leq B$.

Theorem 13.14 (Selection is a 0/1 knapsack). *The selection problem is equivalent to a 0/1 knapsack with values $v_i = -\log(1 - \beta_i/\Omega) > 0$, costs c_i , and capacity B . It is solvable exactly by dynamic programming in $\mathcal{O}(kB)$ time and space, and the value-density greedy rule—admit probes in decreasing order of v_i/c_i —is optimal under the linear relaxation and within a factor $(1 - c_{\max}/B)$ of optimal for the integral problem (Dantzig, 1957; Kellerer et al., 2004; Sahni, 1975).*

Proof. Taking logarithms of the objective, $\log(\Omega \prod_{i \in S} (1 - \beta_i/\Omega)) = \log \Omega - \sum_{i \in S} v_i$, so

minimising the composite residual is maximising $\sum_{i \in S} v_i$ subject to the budget, which is the stated knapsack verbatim. The dynamic program is standard (Cormen et al., 2009); the greedy bound and the fully polynomial approximation scheme are Dantzig (1957) and Sahni (1975). $\square$

Corollary 13.15 (Selection priority). *A probe with residual β_i and cost c_i has selection priority*

$$\varrho_i = \frac{1}{c_i} \log \frac{\Omega}{\Omega - \beta_i},$$

and probes are admitted in decreasing order of ϱ_i .

Definition 13.16 (Probing allocation; gain profile). Suppose instead that a probe may be applied to a graded extent: an *allocation* is a vector $\mathbf{a} = (a_1, \dots, a_k) \in [0, \infty)^k$ with $\sum_i a_i \leq \alpha$ for a budget $\alpha > 0$, where a_i is the effort spent on probe i . The *gain profile* $\gamma_i: [0, \infty) \rightarrow [0, \infty)$ gives the above-floor alignment $\gamma_i(a_i)$ that probe i removes at effort a_i , with $\gamma_i(0) = 0$. The value of an allocation is $G(\mathbf{a}) = \sum_i \gamma_i(a_i)$.

Premise 3 (Diminishing returns within a probe). Each gain profile γ_i is concave and nondecreasing.

Remark 13.17 (This premise is about the environment, not the system). Premise 3 is the one premise in this paper that is a claim about the problem being probed rather than about the structure doing the probing. It says that the second unit of effort spent on a probe removes no more alignment than the first. Where it fails—where a probe has a threshold below which it returns nothing and above which it returns a great deal—Theorem 13.18 does not apply and the discrete selection of Theorem 13.14 is the correct model. We flag the dependence explicitly; every other result in the paper is independent of it.

Theorem 13.18 (Optimal division is water-filling). *Under Premise 3, the value-maximising allocation solves $\max_{\mathbf{a} \geq 0, \sum_i a_i \leq \alpha} \sum_i \gamma_i(a_i)$, and there is a single shadow price $p^* \geq 0$ such that the optimum satisfies*

$$\gamma'_i(a_i^*) = p^* \text{ if } a_i^* > 0, \quad \gamma'_i(0) \leq p^* \text{ if } a_i^* = 0.$$

The system equalises marginal gain across all engaged probes at the price p^ , and drops any probe whose first unit of effort returns less than p^* . The price is nonincreasing in the budget α and nondecreasing in the number and richness of competing probes; if the budget is not exhausted at the optimum then $p^* = 0$.*

Proof. The objective is concave as a sum of concave functions and the feasible set $\{\mathbf{a} \geq 0 : \sum_i a_i \leq \alpha\}$ is compact and convex, so a maximiser exists (Boyd and Vandenberghe, 2004). Form the Lagrangian $L = \sum_i \gamma_i(a_i) - p^*(\sum_i a_i - \alpha) + \sum_i \mu_i a_i$ with $p^* \geq 0, \mu_i \geq 0$. Stationarity gives $\gamma'_i(a_i) - p^* + \mu_i = 0$ with complementary slackness $\mu_i a_i = 0$ and $p^*(\sum_i a_i - \alpha) = 0$ (Kuhn and Tucker, 1951). For $a_i^* > 0, \mu_i = 0$ yields $\gamma'_i(a_i^*) = p^*$; for $a_i^* = 0, \mu_i \geq 0$ yields $\gamma'_i(0) \leq p^*$. Sufficiency of the conditions for a global optimum follows from concavity. Budget monotonicity: relaxing α enlarges the feasible set, so the optimal value is nondecreasing and, being concave in α , its marginal p^* is nonincreasing. Adding a probe or raising a profile tightens the binding budget and raises the level at which margins equalise. If the budget is slack at the optimum, its multiplier is zero. $\square$

Corollary 13.19 (Every probe engaged, at a price). *At the optimum the system spends positive effort in exactly the probes whose marginal gain at zero exceeds p^* , and none in the rest; within engaged probes it holds every marginal gain equal to p^* . Thus all engaged probes are working simultaneously, at graded effort, and p^* is a single readable scalar for how loaded the system is. A probe whose entry margin only marginally exceeds p^* receives a floor-sized allocation—the boundary case of minimal engagement.*

Proof. Immediate from the stationarity conditions of Theorem 13.18: the support of $\mathbf{a}^*$ is $\{i : \gamma'_i(0) > p^*\}$, margins on the support equal p^* , and p^* falls as α rises. $\square$

Remark 13.20 (Selection versus allocation). Theorem 13.14 and Theorem 13.18 are the discrete and continuous faces of one problem. Where a probe is an all-or-nothing invocation, selection is a knapsack and probes take turns. Where a probe admits graded effort and Premise 3 holds, allocation is water-filling and every probe above the price works at once. The knapsack is the relaxation's boundary: a probe dropped at $a_i^* = 0$ is a probe excluded from the knapsack's chosen set.

13.5 Finding and pruning are different operations

The last structural result about probing concerns architecture. We show that locating what bears

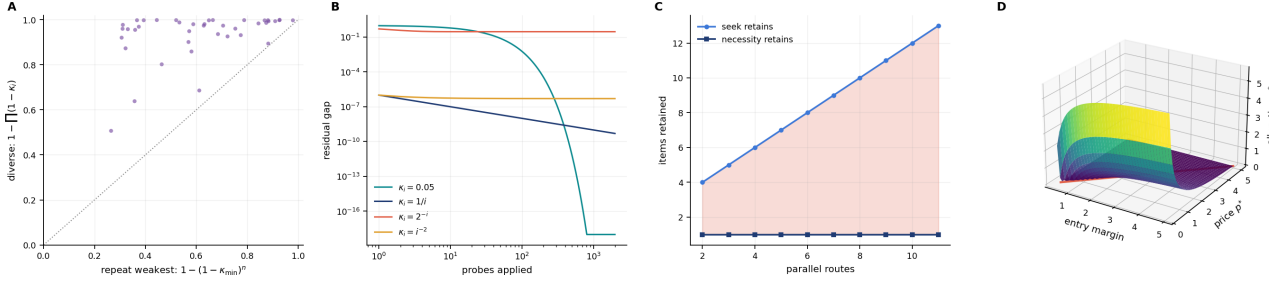


Figure 5: Probes Compound Rather Than Vote, Saturate Only Under a Divergent Sum, and Divide a Budget at a Single Price. **(A)** Composite power of a chain of distinct probes, $1 - \prod_i (1 - \kappa_i)$, on the linear y -axis versus the composite power of repeating its weakest member the same number of times, $1 - (1 - \kappa_{\min})^n$, on the linear x -axis, over 300 instances with the identity diagonal in grey. Every point lies strictly above the diagonal—300 of 300, with no ties—so diversification strictly dominates repetition (Corollary 13.8). The underlying composition law is verified separately on 400 chains to a maximum absolute error of 1.1×10^{-16} (Theorem 13.5). **(B)** Residual gap on a log y -axis versus probes applied on a log x -axis (1 to 2000), for four canonical power sequences. The two with divergent sums—constant $\kappa_i = 0.05$ (teal) and harmonic $\kappa_i = 1/i$ (navy)—drive the residual toward zero; the two with convergent sums—geometric $\kappa_i = 2^{-i}$ (crimson) and inverse-square $\kappa_i = i^{-2}$ (amber)—plateau strictly above it and stay there. At a horizon of 4000 probes all four sequences match the prediction of Corollary 13.7: 0 mismatches. **(C)** Items retained on the linear y -axis versus parallel routes on the linear x -axis (2 to 11), on the diamond witness of Theorem 13.27. The seek operator (steel) climbs linearly from 4 to 13 while nec (navy) stays flat at 1; the shaded region between them is the over-retention a reachability-only mechanism incurs, growing from 3 to 12 items. On the chain witness the complementary identity holds at every tested length: every interior item is necessary and the terminal leaf is reachable yet redundant. Over 60 random graphs the containment $\text{nec} \subseteq \text{reach}$ holds without exception, with a strictly positive gap in 57, and necessity coincides with domination on all 211 non-seed items checked (Corollary 13.28). **(D)** Optimal allocation a_i^* as a three-dimensional surface (viridis) over entry margin $\gamma_i'(0)$ on the x -axis (0.4 to 5.0) and shadow price p^* on the y -axis (0.25 to 5.0), elevation 24° azimuth -58° . The surface rises with margin and falls with price, collapsing to the zero floor along the crimson line where the entry margin meets the price—the sharp dropout boundary of Theorem 13.18. Across 120 convex instances the stationarity conditions hold, the budget is respected, the dropout boundary is sharp, and the price is nonincreasing in the budget and nondecreasing in the probe count, in 120 of 120 cases each.

on a target and determining what may be discarded are distinct computations, so no single mechanism performs both.

Definition 13.21 (Reachability). Seed a frontier with the target’s distinctions. An item is *reachable at distance 0* if it shares a distinction with the target, and at distance $d + 1$ if it is not reachable at distance $\leq d$ and shares a distinction with some item reachable at distance $\leq d$. Write $\text{reach}(x^*)$ for the set of reachable items.

Definition 13.22 (Resolution functional; contribution; necessity). Let $R(x^* | W)$ be a *resolution functional* on retained sets W : a monotone measure of how much of the target’s neighbourhood the retained items recover, increasing as items are added. The canonical choice is $R(x^* | W) = |\text{reach}_W(x^*)|$. The *contribution* of $u \in W$ is

$$\varsigma(u, x^*) = R(x^* | W) - R(x^* | W \setminus \{u\}) \geq 0,$$

and u is *necessary* if $\varsigma > 0$, *purposeless* if $\varsigma = 0$.

Remark 13.23 (The orientation of the functional matters). R must *grow* with resolving power, so that removing a load-bearing item lowers it and contributions are non-negative. A minimum cut of the target against the medium has the wrong orientation for this role: deleting an item deletes edges, which can only shrink a cut, making every contribution non-positive. The reachable-slice measure, or any monotone measure of recovered neighbourhood, has the correct sign.

Proposition 13.24 (Necessity is contained in reachability). $\text{nec}(x^*) \subseteq \text{reach}(x^*)$: *an item the target cannot reach is never necessary. The reverse inclusion fails in general.*

Proof. If $u \notin \text{reach}(x^*)$ then no path of shared distinctions connects u to the target, so removing u deletes only edges incident to u , none of which lies on any path from the target into its reachable neighbourhood. Hence $R(x^* | W \setminus \{u\}) = R(x^* | W)$ and $\varsigma = 0$. For failure of the reverse inclusion, see Theorem 13.27(iii). $\square$

Proposition 13.25 (Necessity is domination). *A non-seed item is necessary for a target exactly when it dominates some reachable item—lies on every path from the target to it. Domination is computable in near-linear time (Lengauer and Tarjan, 1979).*

Proof. If u dominates r , every path from the target to r passes through u , so removing u removes r from the reachable set and strictly lowers R . Conversely if u dominates nothing, then for every reachable r there is a path from the target to r avoiding u , so removing u leaves the reachable set unchanged and $\varsigma = 0$. $\square$

Remark 13.26 (Two things the measurement must get right). Theorem 13.22 compares the resolution of the retained set against that of the set with u removed, and the comparison must be taken over the items *other than* u . Counting u ’s own disappearance from the reachable set gives every reachable item a contribution of at least one and collapses *nec* onto *seek*—which is exactly the degeneracy Theorem 13.27 forbids, and which the validation suite detected as a failure before the measurement was corrected (??).

The seed is a genuine boundary case rather than an artefact. A seed is the target’s only point of entry and is load-bearing by definition; where a seed reaches nothing but itself it dominates no item while remaining necessary. Theorem 13.25 is therefore stated for non-seed items, and a system computing necessity must treat seeds separately rather than by the domination test.

Theorem 13.27 (Separation of finding and pruning). *The operators $\text{seek}(x^*) = \text{reach}(x^*)$ and $\text{nec}(W, x^*) = \{u \in W : \varsigma(u, x^*) > 0\}$ compute distinct quantities and are not reducible to one another by a single pass:*

- (i) *seek is a forward reachability from the target—a growth from a seed—computing what the target touches;*
- (ii) *nec is a backward ablation—a removal test against a retained set—computing what a drop changes;*
- (iii) *the two disagree: there exist items that seek marks reachable and nec marks purposeless, and the discrepancy is recoverable from neither operator’s output alone.*

Proof. (i) and (ii) are the definitions, and their computational shapes are opposite: a seed-and-grow traversal versus a per-item ablation. (iii) Construct a redundant reachable item. Let a target distinction be shared by two items u_1, u_2 , each also joined to a common item r , so both lie on a path from the target to r . Both are in $\text{seek}(x^*)$. But if r is reachable via u_1 alone, removing u_2 leaves R unchanged, so u_2 is reachable

yet purposeless: $u_2 \in \text{seek}$ and $u_2 \notin \text{nec}$. The output of seek—the reachable set—does not record which of its members are redundant, and the output of nec—the necessary set—does not record the reachable-but-redundant items it excluded. Neither output determines the other. $\square$

Corollary 13.28 (No single-mechanism probing). *A mechanism running only seek over-retains: it keeps reachable-but-redundant items, wasting budget. A mechanism running only nec has nothing to ablate against, since ς is defined relative to a retained set. Correct probing requires both, in the order $\text{nec} \circ \text{seek}$.*

Proof. Over-retention is Theorem 13.27(iii). The dependence of nec on a retained set is Theorem 13.22, whose natural argument is $\text{seek}(x^*)$. $\square$

Remark 13.29 (Consequence for retrieval architectures). Retrieval-augmented generation (Lewis et al., 2020) implements seek alone: it fetches a relevant subset per query. Theorem 13.28 identifies what is structurally missing—an operator answering *what may be discarded*—and Theorem 13.27 shows it cannot be obtained by making the retriever better, because it is a different computation. No theorem depends on the comparison.

14 Termination: closure, and the honest decline

Theorem 8.7 showed the system cannot issue a verdict. We now supply what it *can* report, and prove the stopping rule is not a confidence threshold.

Definition 14.1 (Reachable class set). Fix a seed v_0 and a process graph whose available probes may be extended over time. At a given stage, let $\mathcal{C}(v_0)$ denote the set of targets reachable by some admissible propagation of v_0 through the probes invoked so far, partitioned into equivalence classes under endpoint-indistinguishability (Theorem 10.10).

Definition 14.2 (Closure). A determination from seed v_0 is *closed* at a given stage if, for every probe available but not yet invoked, extending by that probe cannot add a new equivalence class to $\mathcal{C}(v_0)$: every further admissible propagation reachable via it terminates in a class already present.

Theorem 14.3 (Closure is strictly stronger than a confidence threshold). *There exist process graphs and thresholds θ such that a propagation’s terminal alignment satisfies a fixed confidence criterion while the determination is not closed in the sense of Theorem 14.2: a not-yet-invoked probe is available whose propagation reaches a target in a distinct equivalence class.*

Proof. Construct a process graph with two disjoint item clusters A and B , each internally dense—many low-cost internal contacts—and each connected to the medium by a single probe, γ_A and γ_B , of equal weight. A propagation via γ_A alone terminates at a representative $x_A^* \in A$ with terminal alignment $a(x_A^*, x_A^*) = \beta/\Omega$, satisfying any fixed threshold $\theta \leq 1$ trivially, since by Theorem 10.8 a propagation’s terminal alignment to its own target is always at the floor. Yet γ_B , not yet invoked, reaches $x_B^* \in B$, in a distinct equivalence class from x_A^* : the two clusters have independent internal structure, so their resting cuts against the medium differ and they are not endpoint-indistinguishable. Hence the confidence criterion is satisfied after γ_A alone while the determination is not closed. $\square$

Remark 14.4 (The failure mode the theorem isolates). Theorem 14.3 formalises a familiar defect of threshold-stopping. A single early source may report high confidence purely because it is internally self-consistent—as any single cluster is, having only committed contacts within itself and one to the medium—while a second, independent source would reach an entirely different conclusion. Closure requires that *every* available further probe be checked to land in an already-reached class before stopping. It is not that the current determination looks confident, but that the space of determinations has stopped producing new destinations.

Theorem 14.5 (Convergent closure or contested closure). *Every determination over a finite probe registry terminates in exactly one of two states:*

- (i) **Convergent closure:** $\mathcal{C}(v_0)$ stabilises to a single equivalence class, and the determination reports that class’s representative;
- (ii) **Contested closure:** $\mathcal{C}(v_0)$ stabilises—no further probe adds a class—but contains more than one class, and the determination reports decline: no single sufficient determination exists, together with the distinct classes found.

No determination over a finite registry runs forever without reaching one of these two states.

Proof. The probe registry available at any call is finite, so the set of propagations reachable from v_0 is finite, and hence $\mathcal{C}(v_0)$ —a partition of a finite set of reachable targets—is finite and can grow by only finitely many steps before every remaining probe has been tried against every currently-reached class. At that point Theorem 14.2 is satisfied by exhaustion, so closure is reached in finitely many steps. If the stabilised $\mathcal{C}(v_0)$ has exactly one class, the determination is convergently closed; if more than one, then by Theorem 10.10 each class is internally endpoint-indistinguishable while the classes disagree with one another, so no single sufficient determination is licensed and decline is reported. The two cases are exhaustive and mutually exclusive on a partition with at least one class, and at least one class exists once any propagation is admissible. $\square$

Corollary 14.6 (Decline is a first-class output). *Contested closure is a correctly-terminating outcome carrying the distinct classes found. It is not a failure to be retried, and by Theorem 8.7 the system could not adjudicate between the classes even if instructed to.*

Proof. Theorem 14.5(ii) establishes contested closure as one of two terminal states of a terminating procedure, and Theorem 8.7 forbids the system from selecting among the classes. $\square$

Remark 14.7 (What contested closure is worth). A determination that ends in contested closure has located precisely where independent probing diverges. That is ordinarily more valuable than either class alone, and is exactly the information a threshold-stopping procedure discards: by Theorem 14.3 such a procedure would have halted at whichever class it reached first and reported it as sufficient. No theorem depends on the reading.

Corollary 14.8 (Anytime discipline). *The correct stopping condition is non-resolution against the floor, not budget exhaustion: a determination whose reachable propagations do not settle the target’s distinctions to within β should descend further or decline, never fabricate. This sharpens the anytime stance (Dean and Boddy, 1988; Zilberstein, 1996) by moving the stopping criterion from elapsed effort to structural closure.*

Proof. By Theorem 3.5 “resolved” is a region with a positive boundary; a target whose reachable

propagations never enter that region is genuinely unresolved by the current structure, and Theorem 14.5 supplies decline as the terminal state for that case. $\square$

Part V

The Separation

15 What a store of local content cannot express

We now prove the paper’s second main result. The accountability verdict of Theorem 10.5 is a global minimum over the item set expressed in local terms, and no system whose answer is a function of stored local content can express it. This covers triple stores, attributed triple stores, and—by Theorem 15.8—corpus-trained predictors.

15.1 Two kinds of question

Definition 15.1 (Retrieval query). Let $\mathcal{G}$ be a finite set of assertions over a vocabulary. A *retrieval query* is a map q assigning to $\mathcal{G}$ a set of tuples $q(\mathcal{G})$, computed by pattern matching and path composition on $\mathcal{G}$ alone.

Definition 15.2 (Admissibility query). Given a contact graph, a seed v_0 , a target x^* , a process Π , and a tolerance ε , the *admissibility query* returns the accountable class $\mathcal{A}(v_0, x^*, \Pi)$ of Theorem 10.5—equivalently the entry $\mathcal{T}(v_0, x^*, \Pi)$ —together with its alignment value.

The difference is not one of expressive convenience. A retrieval query asks *what is recorded*. An admissibility query asks *whether a determination from here to there can be accounted for*, and it has an answer for pairs about which nothing was ever recorded.

15.2 The separation

Unfolding (1), a determination is ε -accountable exactly when $\sigma(v_0, x^*) \leq \beta^* + \varepsilon\Omega$, so at $\varepsilon = 0$ the total weight cancels and the verdict is a comparison of two numbers: a single minimum cut, and the system floor. The first is local to the queried pair. The second is a minimum over *every* item in the system. The theorem below is nothing more than the observation that these two can be moved independently, and that an assertion records neither.

Theorem 15.3 (Retrieval cannot express admissibility). *There are contact graphs Γ_1, Γ_2 on the same item set, with the same contact relation, and a pair (v_0, x^*) , such that*

- (i) $q(\mathcal{G}_1) = q(\mathcal{G}_2)$ for every retrieval query q , where $\mathcal{G}_i$ records the contact relation of Γ_i ; while
- (ii) $\mathcal{A}(v_0, x^*, \Pi) \neq \emptyset$ in one and $\mathcal{A}(v_0, x^*, \Pi) = \emptyset$ in the other, at $\varepsilon = 0$.

Proof. Fix $n \geq 2$ and let $V = \{v_0, x^*, y_1, \dots, y_n, m\}$. In both graphs the edges are $\{v_0, x^*\}$ and $\{u, m\}$ for every $u \in V \setminus \{m\}$. Both are contact graphs, since every item is joined to the medium.

Assign weights as follows. In both, $w(\{v_0, x^*\}) = 1$ and the medium edges at v_0 and at x^* have weight 1. The graphs differ only in the medium edges at the y_i : weight 2 in Γ_1 , weight 1 in Γ_2 .

(i) The edge sets are equal, $E_1 = E_2$, so the recorded contact relations coincide: $\mathcal{G}_1 = \mathcal{G}_2$ as sets of assertions. Any q in the sense of Theorem 15.1 is a function of its argument alone, hence $q(\mathcal{G}_1) = q(\mathcal{G}_2)$. An assertion states that a relation holds and carries no cost, so there is nothing here for retrieval to see.

(ii) Compute both sides of (1). The quantity $\sigma(v_0, x^*)$ is the minimum weight of a cut separating v_0 from x^* . The y_i are joined only to m , so they lie on no v_0 – x^* path and no minimum such cut contains their edges; the cheapest separation places v_0 alone on one side, severing $\{v_0, x^*\}$ and the medium edge at v_0 , at cost 2. Hence $\sigma(v_0, x^*) = 2$ in both graphs, the weights involved being identical.

The system floor is a minimum over all items. By Theorem 2.4, $\sigma(u)$ is the cost of separating u from m . For $u \in \{v_0, x^*\}$ this is 2 in both graphs: isolating v_0 severs its medium edge and $\{v_0, x^*\}$. For $u = y_i$, isolating y_i severs its single medium edge, so $\sigma(y_i) = 2$ in Γ_1 and $\sigma(y_i) = 1$ in Γ_2 . Therefore

$$\beta_1^* = \min(2, 2, 2, \dots, 2) = 2, \quad \beta_2^* = \min(2, 2, 1, \dots, 1) = 1.$$

Applying (1) at $\varepsilon = 0$: in Γ_1 we have $\sigma(v_0, x^*) = 2 \leq 2 = \beta_1^*$, so the determination is 0-accountable and $\mathcal{A} \neq \emptyset$; in Γ_2 we have $\sigma(v_0, x^*) = 2 > 1 = \beta_2^*$, so no determination from v_0 to x^* satisfies (1) and $\mathcal{A} = \emptyset$. Both clauses hold. $\square$

Remark 15.4 (What the separation turns on). The construction turns on a single asymmetry. The

queried quantity $\sigma(v_0, x^*)$ depends on the weights along v_0 – x^* separations; the threshold β^* depends on a minimum over every item in the system, including items lying on no path between the two and, from the point of view of the query, entirely irrelevant. Editing a weight at the y_i changes the verdict about (v_0, x^*) without changing anything a pattern-matcher could match against. So *accountability is a global property expressed in local terms*, and a store records only the local terms. This is sharper than saying weights are missing: one may store weights as attributes and still not recover β^* , which is a minimum over items of a minimum over subsets, and neither minimisation is a pattern.

Corollary 15.5 (Attributes do not close the gap). *Extending Theorem 15.1 to allow numeric attributes on assertions and arithmetic comparison in patterns does not make admissibility expressible.*

Proof. By Theorem 15.3 and Theorem 15.4 the verdict depends on β^* , a minimum over items of a minimum over subsets. Conjunctive queries with regular path expressions have polynomial data complexity and are evaluated by matching bounded patterns (Angles et al., 2017; Barceló Baeza, 2013; Chandra and Merlin, 1977; Pérez et al., 2009); the fragment contains no operator forming a minimum over subsets of the domain, and adding comparison of stored attributes introduces none. The obstruction is expressibility, not hardness: a minimum cut is computable in polynomial time (Ford and Fulkerson, 1956; Stoer and Wagner, 1997), so the quantity is cheap to compute and impossible to phrase. $\square$

15.3 The separation extends to corpus-trained predictors

We now extend the result past stores of assertions to the object of Section 7.1. The extension requires one further definition and turns on the same fact.

Definition 15.6 (Corpus-determined map). Let $\mathcal{K}$ be a finite corpus—any finite body of recorded local content. A map D from queries to determinations is *corpus-determined by $\mathcal{K}$* if D is a function of $\mathcal{K}$ alone: there is a procedure Λ with $D = \Lambda(\mathcal{K})$, and D ’s outputs do not depend on anything not recorded in $\mathcal{K}$.

Remark 15.7 (The scope of the definition). Theorem 15.6 is deliberately indifferent to how Λ

works. It covers a lookup table, a compression of $\mathcal{K}$, an interpolator fitted to $\mathcal{K}$, and a predictor whose parameters were optimised against $\mathcal{K}$; it covers deterministic and randomised Λ alike, provided the randomness is independent of the graph being queried. What it requires is only that no information enters D except through $\mathcal{K}$.

Theorem 15.8 (Corpus-determined maps cannot express admissibility). *Let $D = \Lambda(\mathcal{K})$ be corpus-determined, and suppose $\mathcal{K}$ records the contact relation of a contact graph—i.e. which items contact which. Then there exist contact graphs Γ_1, Γ_2 inducing the same $\mathcal{K}$ and a pair (v_0, x^*) such that D returns the same determination on both while the admissibility verdicts differ. Hence no corpus-determined map expresses admissibility.*

Proof. Take Γ_1, Γ_2 as constructed in the proof of Theorem 15.3. By clause (i) of that proof the two graphs have identical contact relations, so they induce the same corpus: $\mathcal{K}_1 = \mathcal{K}_2 =: \mathcal{K}$. By Theorem 15.6, $D = \Lambda(\mathcal{K})$ is a function of $\mathcal{K}$ alone, so D is the same map in both cases and returns the same determination on (v_0, x^*) . By clause (ii) the admissibility verdicts differ at $\varepsilon = 0$: accountable in one graph, not in the other. A single output cannot equal two different verdicts, so D does not express admissibility. $\square$

Corollary 15.9 (Enlarging the corpus does not help). *Enlarging $\mathcal{K}$ to record numeric attributes on contacts, or any further finite body of local content about individual items and their contacts, leaves Theorem 15.8 intact.*

Proof. By Theorem 15.4 the verdict depends on β^* , a minimum over all items of a minimum over subsets. A corpus of local content records values attached to items and contacts; forming a minimum over subsets of the domain is not such a value, and by Theorem 15.6 D has access to nothing else. The argument of Theorem 15.5 applies verbatim with Λ in place of the query evaluator. $\square$

Theorem 15.10 (What the pair expresses). *The directional pair computes admissibility. For a pair (v_0, x^*) and a process Π , deciding (1) requires one minimum cut between v_0 and x^* and one minimum over items of separation costs, both exactly computable by maximum flow, and returns the accountable class by Theorem 10.15.*

Proof. By Theorem 10.5 the verdict is the comparison $\sigma(v_0, x^*) \leq \beta^* + \varepsilon\Omega$. The left side is a

single v_0 – x^* minimum cut; the right side requires β^* , which by Theorem 10.4 is $\min_v \sigma(v)$, a minimum of $|V| - 1$ minimum cuts; and $\Omega = w(E)$ is a sum. All are exactly computable (Edmonds and Karp, 1972; Ford and Fulkerson, 1956; Stoer and Wagner, 1997). The class is then returned by Theorem 10.15. $\square$

Corollary 15.11 (The separation, stated). *There is a class of question—admissibility—that is polynomial-time computable from a contact graph, that the directional pair answers (Theorem 15.10), and that no store of assertions (Theorem 15.3), no store of attributed assertions (Theorem 15.5), and no corpus-determined map (Theorem 15.8) can express.*

Remark 15.12 (Where the comparison flatters and where it undersells). Two things must be said against the corollary. It *flatters* the construction, because the cost of building the contact graph—extracting the term map from sources—is excluded from the complexity count, whereas a corpus-determined map pays its construction cost once and amortises it over every later query. It *undersells*, because a corpus-determined map answers only about the content someone recorded, whereas Theorem 15.10 holds uniformly over every pair of items in the graph, including pairs about which nothing was recorded. The honest summary is that the two are not competing at the same task: one returns determinations from a finite recorded body, the other computes verdicts over the whole induced structure.

16 The master equivalence

We close the theory by showing every structure above is the same fact.

Theorem 16.1 (Master equivalence). *For a contact graph over a non-completable medium, the following are equivalent:*

- (i) *there is an identifiable item—a proper part told from the rest at positive cost;*
- (ii) *the floor is positive, $\beta > 0$ (Theorem 3.4);*
- (iii) *individuation is by negation without selector (Theorem 4.2); identity is the invariant minimum cut and never a point (Theorems 5.2 and 5.3); the invariant is not extractable by its bearer (Theorem 8.1); processes are*

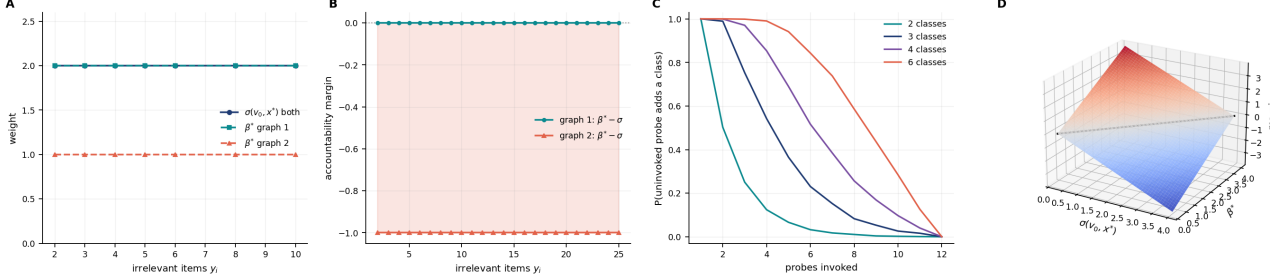


Figure 6: The Verdict Is a Global Quantity, and the Witness Is Invisible to Every Local One. (A) The witness pair of Theorem 15.3 as the number of irrelevant items y_i grows along the linear x -axis (2 to 10), weights on the linear y -axis. The pair alignment $\sigma(v_0, x^*)$ is identical in the two graphs (navy, filled and open markers coincident at exactly 2.0), while the system floors separate: 2.0 in graph 1 (teal) against 1.0 in graph 2 (crimson). The two graphs have identical contact relations at every size—the same assertion set, hence identical under every retrieval query—yet the verdicts differ at $\varepsilon = 0$. (B) The accountability margin $\beta^* - \sigma(v_0, x^*)$ on the linear y -axis, swept to 25 irrelevant items on the linear x -axis. Graph 1 sits at exactly 0 (accountable, teal) and graph 2 at exactly -1 (not accountable, crimson), the two shaded to the zero reference and both flat: adding items that lie on no path between the queried pair moves the verdict without moving anything local to it. This isolates the asymmetry of Remark 15.4—the queried quantity is local and equal, the threshold is global and different. A bounded pattern confined to the queried pair’s neighbourhood is fooled in 200 of 200 instances (Corollary 15.5), and none of 500 corpus-determined maps spanning five families separates the two graphs (Theorem 15.8). (C) Probability that an uninvoked probe still reaches a new equivalence class, on the linear y -axis (0 to 1), versus probes already invoked on the linear x -axis (1 to 12), for registries admitting two, three, four, and six classes, 4000 draws per point. A fixed confidence criterion is satisfied at the first probe while this probability remains near unity, and it decays only as the registry is exhausted—the content of Theorem 14.3 (Closure Is Strictly Stronger). Over 300 runs to closure, 169 terminated in convergent closure and 131 in contested closure, with none running on (Theorem 14.5). (D) The verdict surface: the accountability margin as a three-dimensional surface (coolwarm) over the pair alignment $\sigma(v_0, x^*)$ on the x -axis and the system floor β^* on the y -axis, both 0.2 to 4.0, at $\varepsilon = 0$, elevation 22° azimuth -60° . The black line is the decision boundary $\sigma = \beta^*$, above which a determination is accountable and below which it is not. The two witness graphs sit on opposite sides of this boundary while agreeing on every quantity a store of assertions records; the pair decides the verdict correctly on 200 of 200 random instances and separates the witness that no corpus-determined map separates (Theorem 15.10).

monotone non-returning residue chains (Theorems 6.2 and 6.4); every act of individuation blunts its instrument (Theorem 7.5); and the catalogue is the propagation table at rest (Theorem 11.3);

- (iv) *the perfect repeatable traceless instrument—an individuation leaving no residue and no trace in the instrument—does not exist (Theorem 7.6).*

Proof. (i) $\Rightarrow$ (ii): an identifiable proper part has positive separation cost by Theorems 3.3 and 3.4. (ii) $\Rightarrow$ (iii): each listed structure is proved from $\beta > 0$ in the theorem cited. (iii) $\Rightarrow$ (iv): Theorem 7.6 is among the consequences, being proved from Theorem 3.4 and Theorem 6.4, both listed. (iv) $\Rightarrow$ (i): if no traceless instrument exists then every individuation costs more than zero, i.e. some thing is told from the rest at positive cost—an identifiable item. The cycle closes. $\square$

Remark 16.2 (One fact, many faces). Theorem 16.1 says the paper has one premise and many restatements of its consequence. Telling a thing from the rest of a whole that cannot be finished costs a strictly positive irreducible residue. That residue is at once the thing’s identity (Theorem 5.3), the cause of the next determination (Theorem 6.2), the trace deposited in the instrument (Theorem 7.5), the reason no determination is a point (Theorem 3.5), and—read at rest versus under process—the reason a catalogue and a propagation machinery are one object (Theorem 11.3).

Part VI

Architecture, Validation, and Scope

17 The blueprint

We convert the theory into commitments a running system must honour. Each is stated as a checkable predicate together with the theorem that makes the predicate the right one. These are design commitments, honourable by construction, not empirical hypotheses about any existing system.

Invariant 17.1 (Conserved invariant). The system computes, from its own structure alone, the

family of separation costs $(\sigma(v))_v$ and their realising cuts, and this family is preserved under every relabelling of items.

Predicate: for a random relabelling ϕ , the multiset $\{\sigma(v)\}$ is unchanged and each resting cut maps onto a resting cut.

Certified by: Theorem 5.2, Theorem 5.5.

Invariant 17.2 (Never-resetting record). Every committed act increments a monotone counter that no operation decrements, and two states with equal configuration and unequal counters are distinct states.

Predicate: no code path decreases the counter; “undo” is implemented as a compensating commit that increments it.

Certified by: Theorem 6.4, Theorem 6.5.

Invariant 17.3 (Determination by search, not by lookup). Every determination is a freshly computed cut against the current graph. No determination is returned from a cache of prior determinations.

Predicate: the determination path contains no read of a stored answer; every answer’s provenance is a set of cuts computed at the current record.

Certified by: Theorem 11.13, Theorem 7.10.

Invariant 17.4 (Deposit on every propagation). Every process-side propagation commits its residue into the graph before its determination is returned.

Predicate: the record after a determination strictly exceeds the record before; the uncommitted capacity of the items involved strictly decreases.

Certified by: Theorem 7.5, Theorem 11.8.

Invariant 17.5 (Accretive update). Structural change adds items, contacts, and committed values; no update leaves a unit open, and no update is implemented as teardown followed by rebuild.

Predicate: for every unit and every intermediate stage of an update, a resting cut is attained.

Certified by: Theorem 9.7, Theorem 9.9, Theorem 9.8.

Invariant 17.6 (No system verdict). The system reports determinations, classes, and declines; it does not report success or failure.

Predicate: the vocabulary exposed to consumers contains no comparison of an achieved determination to an expected one.

Certified by: Theorem 8.7, Theorem 8.6, Theorem 14.5.

Proposition 17.7 (The invariants are independent). *No invariant of Theorem 17.1–Theorem 17.6 follows from the conjunction of the others: for each, there is a system satisfying the remaining five and violating it.*

Proof. For each invariant we exhibit the violating system. Violating Theorem 17.1: a system storing labels as identity, which satisfies the others but fails relabelling-invariance by Theorem 5.2. Violating Theorem 17.2: a system with a resettable counter, which may still compute cuts freshly and accrete. Violating Theorem 17.3: a system caching determinations, which may still keep a monotone record. Violating Theorem 17.4: a system computing cuts freshly but discarding the residue, which keeps a record advanced by other means. Violating Theorem 17.5: a system that rebuilds on migration but otherwise behaves correctly. Violating Theorem 17.6: a system that adjudicates between classes at contested closure while honouring the other five. $\square$

Remark 17.8 (What a violation costs). The invariants are not stylistic. A system violating Theorem 17.3 exhibits Theorem 7.10; one violating Theorem 17.4 occupies the position of Theorem 7.8; one violating Theorem 17.5 silently replaces one individual with another (Theorem 9.8) while appearing to preserve it, since χ is unchanged; one violating Theorem 17.6 presents contested closure as a determination, discarding the very information Theorem 14.5(ii) identifies as valuable.

18 Algorithms

We state the two procedures the construction requires, with their complexity.

Algorithm 1 Construction to closure

Require: sources $\mathcal{U}$; constructors $C_1, \dots, C_r$ with $r \geq 3$ of distinct kind
Ensure: induced graph Γ_τ , or a report of contested closure

```

1:  $\mathcal{C} \leftarrow \emptyset$ ;  $\tau \leftarrow$  empty map
2: repeat
3:   for each constructor  $C_j$  not yet exhausted
4:     do
5:        $\tau_j \leftarrow C_j(\mathcal{U})$ 
6:        $c \leftarrow$  class of  $\tau_j$  under endpoint-
7:         indistinguishability
8:       if  $c \notin \mathcal{C}$  then
9:          $\mathcal{C} \leftarrow \mathcal{C} \cup \{c\}$ 
10:      end if
11:   end for
12: until no constructor adds a class to  $\mathcal{C}$   $\triangleright$ 
13:   closure, Theorem 14.2
14: if  $|\mathcal{C}| = 1$  then
15:   return  $\Gamma_\tau$  built from the single class  $\triangleright$ 
16:   Theorem 14.5(i)
17: else
18:   return DECLINE( $\mathcal{C}$ )  $\triangleright$  Theorem 14.5(ii)
19: end if
```

Proposition 18.1 (Construction terminates). *Algorithm 1 terminates in finitely many rounds, and its output is either a single-class graph or a contested-closure report.*

Proof. The constructor set is finite and $\mathcal{C}$ is a partition of a finite set of reachable classes, so $\mathcal{C}$ can grow only finitely often; once no constructor adds a class the loop exits. The dichotomy is Theorem 14.5. $\square$

Algorithm 2 Determination by the directional pair

Require: graph Γ ; seed v_0 ; target x^* ; probes $\gamma_1, \dots, \gamma_k$; budget α ; tolerance ε

Ensure: accountable class or decline; Γ advanced

```

1:  $R \leftarrow \text{seek}(x^*)$   $\triangleright$  reachability,  $\mathcal{O}(|V| + |E|)$ 
2:  $N \leftarrow \text{nec}(R, x^*)$   $\triangleright$  domination, near-linear
   (Lengauer and Tarjan, 1979)
3:  $\mathbf{a}^* \leftarrow \text{water-filling over } N \text{ at price } p^*$   $\triangleright$ 
   Theorem 13.18
4:  $\mathcal{C} \leftarrow \emptyset$ 
5: repeat
6:   invoke each  $\gamma_i$  with  $a_i^* > 0$ ; commit
   residues into  $\Gamma$   $\triangleright$  Theorem 17.4
7:    $\mathcal{C} \leftarrow \mathcal{C} \cup \{\text{classes reached}\}$ 
8: until no uninvoked probe adds a class  $\triangleright$ 
   closure, Theorem 14.2
9:  $\beta^* \leftarrow \min_v \sigma(v)$ ;  $s \leftarrow \sigma(v_0, x^*)$ 
10: if  $s \leq \beta^* + \varepsilon\Omega$  and  $|\mathcal{C}| = 1$  then
11:   return the single class  $\triangleright$  Theorem 10.5
12: else
13:   return DECLINE( $\mathcal{C}$ )
14: end if
```

Theorem 18.2 (Determination cost). *Algorithm 2 performs $\mathcal{O}(|V|)$ minimum-cut computations, one reachability traversal, one dominator computation, one convex allocation, and at most $|\mathcal{C}| \cdot k$ probe invocations. All components are exactly computable; no step requires an index over stored determinations and no step reads a cached answer.*

Proof. The system floor requires $|V| - 1$ minimum cuts by Theorem 10.4, and the pair alignment one more; each is exact by maximum flow (Edmonds and Karp, 1972; Ford and Fulkerson, 1956; Stoer and Wagner, 1997). Reachability is linear (Tarjan, 1972) and domination near-linear (Lengauer and Tarjan, 1979). The allocation is the convex program of Theorem 13.18, solved by bisection on the scalar price. The probe bound is Theorem 14.5: closure is reached once every probe has been tried against every reached class. Absence of cached reads is Theorem 17.3. $\square$

Remark 18.3 (Where the cost actually sits). Theorem 18.2 counts graph operations and excludes the cost of building the graph, exactly as Theorem 15.12 concedes. On a graph already built, determination is cheap: minimum cuts are polynomial and the remaining steps are near-linear. The expense is construction, and Algorithm 1 makes

it explicit rather than hiding it in an amortised training procedure.

19 Discussion

19.1 Relation to existing practice

Knowledge graphs and query languages.

The construction takes from the knowledge-graph tradition (Hogan et al., 2021; Smith et al., 2007) the idea of a structured medium over which questions are asked, and departs from it in two places. Nodes are not stored records: identity is a computed cut (Theorem 5.3), which is why the identifier-stability and cross-reference problems that attend stored identifiers (McMurry et al., 2017; Wren, 2008) do not arise—there is no identifier to become stale. Edges are not curated assertions: the causal relation is a product of a propagation (Theorem 10.17). And the query semantics differ in kind, not degree: Theorem 15.3 shows admissibility lies outside the expressive power of conjunctive queries with regular path expressions (Angles et al., 2017; Pérez et al., 2009), for the structural reason that the verdict depends on a minimum over the entire item set while pattern matching forms no such minimum.

Retrieval augmentation. Retrieval-augmented generation (Lewis et al., 2020) is, in the vocabulary of Section 13, an implementation of seek alone. Theorem 13.28 identifies the missing operator and Theorem 13.27 shows it is not obtainable by improving the retriever. Separately, Theorem 7.8 applies to the generating component regardless of what is retrieved: supplying material to a determiner does not make the determiner deposit.

Long-context architectures. Efficient long-context methods (Tay et al., 2022; Vaswani et al., 2017) lower the marginal cost of carrying more material but do not change what is carried, and the observed degradation of attention over long inputs (Liu et al., 2024) is a symptom of carrying the wrong thing. Theorem 12.11 states the alternative: carry committed structure, which retains the links (Theorem 6.6(ii)), and regenerate material on demand. The two approaches compose rather than compete.

Hallucination. The literature treats confident incorrect output as a quality defect to be reduced

(Bender et al., 2021; Ji et al., 2023). Theorem 14.3 locates a structural cause: a determiner stopping on confidence halts at the first internally-consistent cluster it reaches, and no amount of calibration converts a confidence threshold into a closure test, because closure quantifies over probes not yet invoked. Theorem 14.5 supplies the alternative terminal state.

Provenance and reproducibility. The relocation of reproducibility from results to protocol (Theorem 11.15) is the provenance stance (Davidson and Freire, 2008; Moreau and Groth, 2013) made structural: by Theorem 11.14 repeated determinations need not agree, so what must be recorded is the protocol and the committed structure, not the determination. This aligns the computational case with the experimental one (Open Science Collaboration, 2015; Peng, 2011; Sandve et al., 2013).

Actor and dataflow models. Probes reading and committing over a shared medium resemble actor and dataflow models (Hewitt et al., 1973; Kahn, 1974; Lee and Parks, 1995), and the monotone record is a logical clock (Lamport, 1978). The departure is that the dependency structure is not declared: by Theorem 10.17 the causal edge set is induced by a propagation rather than given to a scheduler.

19.2 What the development assumes and does not supply

The one input the calculus does not derive is the term map (Theorem 12.1): which distinctions each source draws. Everything downstream is forced—given the graph, the floor, the invariant, the record, the table, the directional identity, the allocation, and the separation are all theorems (Theorem 12.3). The term map is a modelling choice whose quality bounds the resolution of the result but never its soundness (Theorem 12.4), and whose quality is measurable without ground truth (Theorem 12.7). Sharpening it is the natural axis of further work, and is the single point at which a learned component enters, bounded to the task of drawing distinctions.

One further premise is environmental rather than structural: Premise 3, the concavity of gain profiles, on which Theorem 13.18 alone depends. Where it fails, the discrete selection of Theorem 13.14 governs instead. We flag the depen-

dence because it is the only place in the paper where a result rests on a claim about the problem rather than about the structure.

19.3 Honest limitations

1. **Construction cost is excluded.** Every complexity figure counts graph operations on a graph already built (Theorem 18.3, Theorem 15.12). A corpus-determined map amortises its construction over all later queries; the present construction pays per source read. This is a real asymmetry and the honest comparison is not that one dominates but that they answer different questions.
2. **The self-verification result is conditional.** Theorem 8.3 is exactly as strong as its closure hypothesis (Theorem 8.4). The unconditional result is Theorem 8.1, which is weaker: it says the invariant is not extractable as a value, not that self-verification is directly contradictory.
3. **Necessity is domination, not reachability, and seeds are separate.** Theorem 13.24 gives only the inclusion, and Theorem 13.27(iii) exhibits its strictness. The exact criterion is Theorem 13.25, stated for non-seed items; seeds are load-bearing by definition and must be handled apart (Theorem 13.26). A system computing reachability and calling it necessity over-retains, and one that mismeasures the contribution collapses the two operators entirely.
4. **The floor readout is monotone, not calibrated.** Theorem 12.8 states this, and Figure 4C makes it concrete: the degenerate family of term maps—every source drawing identical distinctions—induces among the highest realised floors while discriminating worst. A rising floor indicates refinement is occurring; a high floor is not by itself evidence of a good term map, and optimising it directly would be an error.
5. **Two premises, and where they fail.** Premise 1 and Premise 2 hold for essentially every realised system with bounded memory and an open-ended task. Where an agent is effectively unbounded or its medium is genuinely completable—a closed finite world it can exhaust—the floor may vanish, residues can reach zero, determinations become points, and the development is not compelled. In that degenerate regime a stateless determiner

is not occupying a forbidden position, because the forbidden position exists only when $\beta > 0$. The interest of the account is exactly that realised systems are not in that regime.

19.4 The picture in one statement

Telling a thing from the rest of a whole that cannot be finished costs a strictly positive, irreducible residue, and deposits it into the instrument as much as into the sample. An instrument that claims otherwise claims a position the floor forbids. What exists instead is a single table: read at rest, it is a catalogue of invariant cells; read under process, it returns cells of the same kind that rest cannot produce. The residue of each process flows back into the catalogue as committed structure, which is where the instrument blunts, and which is why the catalogue is the process’s identity. An individual is that catalogue’s conserved cut structure together with its unrepealable record: parts may be wholly replaced without touching it, provided nothing is ever left unterminated; reconstruction to identical specification yields a new individual; and no structure can verify its own identity, so identity is what consuming systems individuate it as, and they may correctly disagree.

20 Conclusion

We set out to describe a pairing of a generative determiner with a structured medium, and found that the pairing is not a composition of two systems.

From a single premise—that the whole against which anything is distinguished cannot be finished—we derived a strictly positive floor on the cost of every act of individuation (Theorem 3.4), and then the fact that gives the floor its force: individuation is symmetric, depositing its residue into the instrument exactly as into the sample (Theorem 7.2, Theorem 7.5). The perfect repeatable traceless instrument is the object the floor excludes (Theorem 7.6), and a determiner that is unchanged by determining asserts it is that object (Theorem 7.8). Its two visible symptoms—answering as a state it has left (Theorem 7.10), and converting its own prior propagation into content whose causal links are gone (Theorem 7.11)—are consequences, not incidental defects.

The remedy is a direction rather than a component. The causal propagation table evaluated

at the trivial process is a catalogue of resting cells, and evaluated at a nontrivial process returns objects of the same type computed under propagation (Theorem 11.3). A structured medium and a generative determiner are therefore one map at two settings of its process argument (Theorem 11.4), and the residue of every process-side propagation is committed into the medium (Theorem 11.8), which is where the instrument blunts and which is the precise sense in which the medium is the determiner’s identity (Theorem 11.9). The pair then has neither symptom: nothing is re-presented (Theorem 11.12) and nothing is stale (Theorem 11.13), and repeated determinations correctly need not agree (Theorem 11.14).

Identity itself turned out to require a pair: the conserved, non-local, relabelling-invariant cut structure together with the unrepealable record (Theorem 9.3). The operative criterion is continuity of termination rather than continuity of parts (Theorem 9.7), which adjudicates gradual replacement, reconstruction, and copying, and yields two engineering commitments—no snapshot-restore, accretive migration only (Theorem 9.8, Theorem 9.9). No structure verifies its own identity (Theorem 8.1, Theorem 8.3), so identity is consumer-relative (Theorem 8.6) and the system issues no verdict (Theorem 8.7).

As a working architecture the pair takes one declared input—the term map—and is correct for any value of it (Theorem 12.3), so a fallible constructor coarsens resolution without corrupting soundness (Theorem 12.4) and its quality is readable from the realised floor without ground truth (Theorem 12.7). Probes over the medium compose multiplicatively rather than by vote (Theorem 13.5), require no fewer than three mutually supporting members (Theorem 13.11), divide a finite budget by water-filling at a single shadow price (Theorem 13.18), and must be paired with a distinct pruning operator that no single mechanism supplies (Theorem 13.27). Determination stops on closure, which is strictly stronger than any confidence threshold (Theorem 14.3), and reports contested closure as a first-class outcome rather than resolving it (Theorem 14.5).

Finally, the pair answers a class of question that no store of local content can express. Accountability is a global minimum over the item set stated in local terms, and the verdict is therefore invisible to pattern matching over assertions (Theorem 15.3), to attributed assertions (The-

orem 15.5), and to any map determined by a finite corpus (Theorem 15.8). The quantity is polynomial-time computable and inexpressible, which is an unusual combination and the reason the gap has persisted (Theorem 15.11).

The construction cost is real and we have not hidden it: building the medium is work that a corpus-determined map amortises and this one pays per source. What is bought is a determiner that has an identity, that cannot be stale, that deposits what it learns into the structure it reasons over, that knows when it has not resolved, and that can say so.

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